



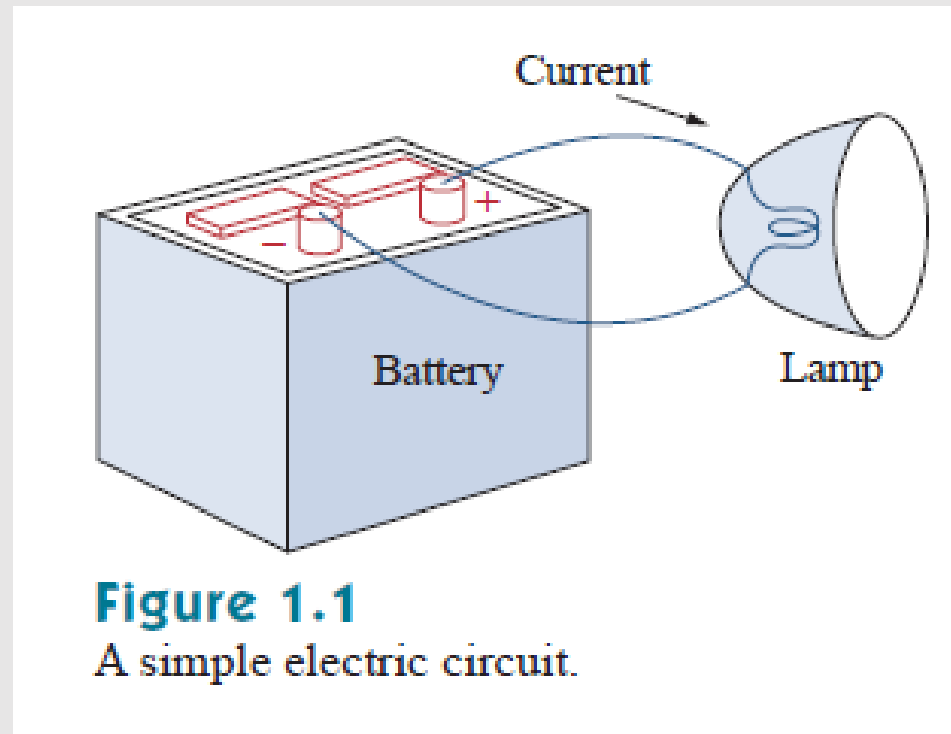
Department of EEE, DUET

Lecture 1: Basic Concepts of Electrical Circuits

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Electrical Circuit

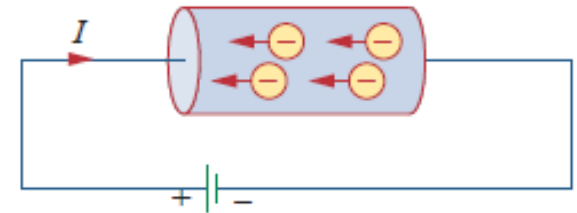
- ❑ An electrical circuit is a interconnection of different electrical elements



Charge and Current

- ❑ Charge is an electrical property of the atomic particles of which matter consists, measured in coulombs (C).
- ❑ The law of conservation of charge states that charge can neither be created nor destroyed, only transferred. Thus the algebraic sum of the electric charges in a system does not change.
- ❑ Electric current is the time rate of change of charge, measured in amperes (A).

Charge and Current



Battery

Figure

Electric current due to flow of electronic charge in a conductor.

Mathematically, the relationship between current i , charge q , and time t is

$$i \triangleq \frac{dq}{dt} \quad (1.1)$$

where current is measured in amperes (A), and

$$1 \text{ ampere} = 1 \text{ coulomb/second}$$

The charge transferred between time t_0 and t is obtained by integrating both sides of Eq. (1.1). We obtain

$$Q \triangleq \int_{t_0}^t i \, dt \quad (1.2)$$

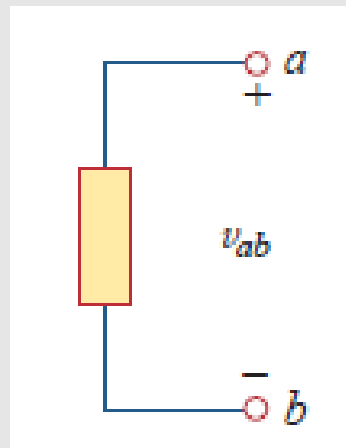
Voltage

- Voltage (or potential difference) is the energy required to move a unit charge through an element, measured in volts (V).

$$v_{ab} \triangleq \frac{dw}{dq}$$

where w is energy in joules (J) and q is charge in coulombs (C). The voltage v_{ab} or simply v is measured in volts (V)

1 volt 1 joule/coulomb 1 newton-meter/coulomb



Power

Power is the time rate of expending or absorbing energy, measured in watts (W).

We write this relationship as

$$p \triangleq \frac{dw}{dt} \quad (1.5)$$

where p is power in watts (W), w is energy in joules (J), and t is time in seconds (s). From Eqs. (1.1), (1.3), and (1.5), it follows that

$$p = \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt} = vi \quad (1.6)$$

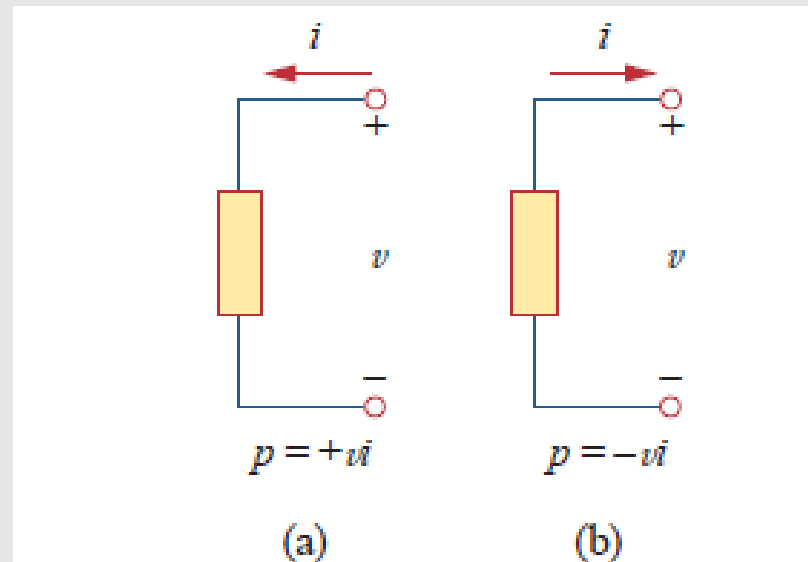
or

$$p = vi \quad (1.7)$$

- ❑ If the power has a positive sign, power is being delivered to or absorbed by the element.
- ❑ If, on the other hand, the power has a negative sign, power is being supplied by the element.

Power: Passive Sign Convention

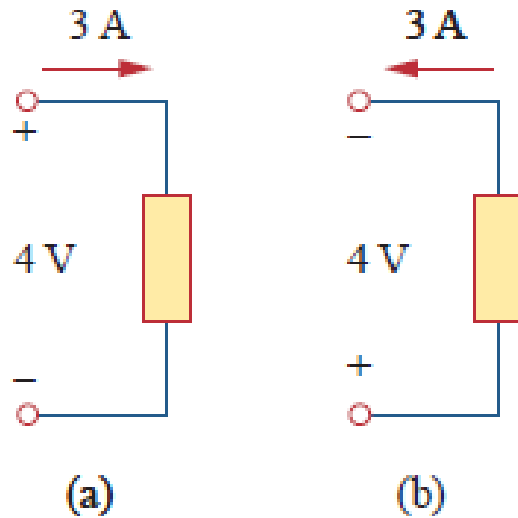
- Passive sign convention is satisfied when the current enters through the positive terminal of an element and $p = +vi$. If the current enters through the negative terminal, $p = -vi$



Figure

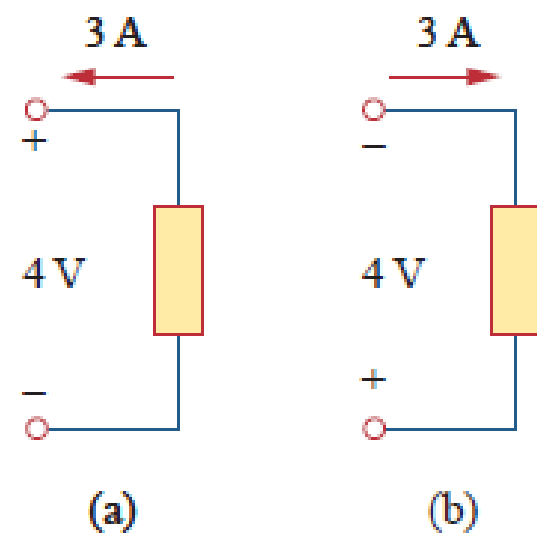
Reference polarities for power using the passive sign convention: (a) absorbing power, (b) supplying power.

Power: Passive Sign Convention



Figure

Two cases of an element with an absorbing power of 12 W: (a) $p = 4 \times 3 = 12$ W, (b) $p = 4 \times 3 = 12$ W.



Figure

Two cases of an element with a supplying power of 12 W: (a) $p = -4 \times 3 = -12$ W, (b) $p = -4 \times 3 = -12$ W.

Energy

- Energy is the capacity to do work, measured in joules (J).

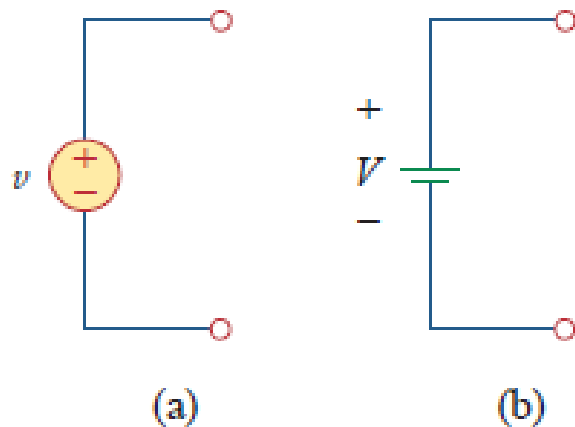
$$w = \int_{t_0}^t p \, dt = \int_{t_0}^t vi \, dt$$

- The electric power utility companies measure energy in watt-hours (Wh), where

$$1 \text{ Wh} = 3,600 \text{ J}$$

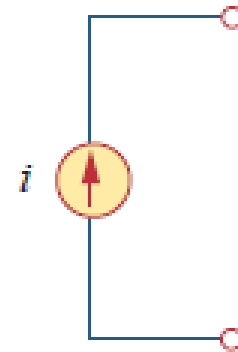
Types of Voltage/Current Sources

- ❑ Independent Source: An ideal independent source is an active element that provides a specified voltage or current that is completely independent of other circuit elements.



Figure

Symbols for independent voltage sources: (a) used for constant or time-varying voltage, (b) used for constant voltage (dc).



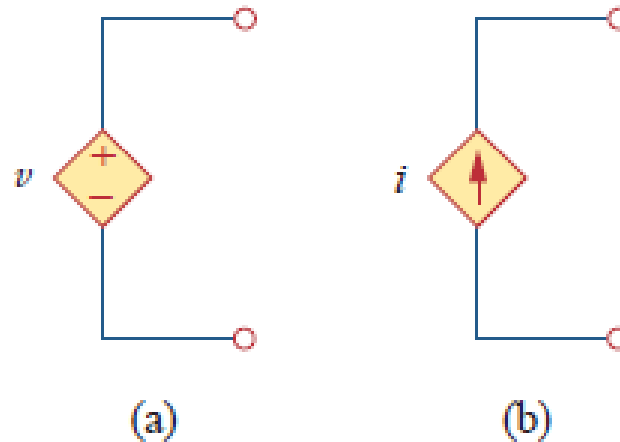
Figure

Symbol for independent current source.

Types of Voltage/Current Sources

❑ **Dependent Source:** An ideal dependent (or controlled) source is an active element in which the source quantity is controlled by another voltage or current. Four (04) types of dependent source are:

- 1) A voltage-controlled voltage source (VCVS)
- 2) A current-controlled voltage source (CCVS)
- 3) A voltage-controlled current source (VCCS)
- 4) A current-controlled current source (CCCS)



Figure

Symbols for: (a) dependent voltage source, (b) dependent current source.

Ohm's Law

Ohm's law states that the voltage v across a resistor is directly proportional to the current i flowing through the resistor.

That is,

$$v \propto i \quad (2.2)$$

Ohm defined the constant of proportionality for a resistor to be the resistance, R . (The resistance is a material property which can change if the internal or external conditions of the element are altered, e.g., if there are changes in the temperature.) Thus, Eq. (2.2) becomes

$$v = iR \quad (2.3)$$

which is the mathematical form of Ohm's law. R in Eq. (2.3) is measured in the unit of ohms, designated Ω . Thus,

The **resistance** R of an element denotes its ability to resist the flow of electric current; it is measured in ohms (Ω).

Ohm's Law

Materials in general have a characteristic behavior of resisting the flow of electric charge. This physical property, or ability to resist current, is known as *resistance* and is represented by the symbol R . The resistance of any material with a uniform cross-sectional area A depends on A and its length ℓ , as shown in Fig. 2.1(a). We can represent resistance (as measured in the laboratory), in mathematical form,

$$R = \rho \frac{\ell}{A}$$

where ρ is known as the *resistivity* of the material in ohm-meters. Good conductors, such as copper and aluminum, have low resistivities

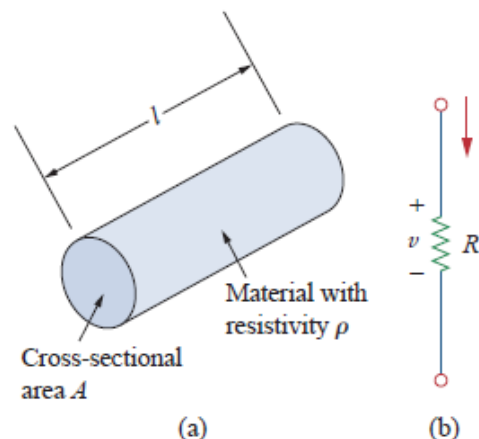


Figure 2.1

(a) Resistor, (b) Circuit symbol for resistance.

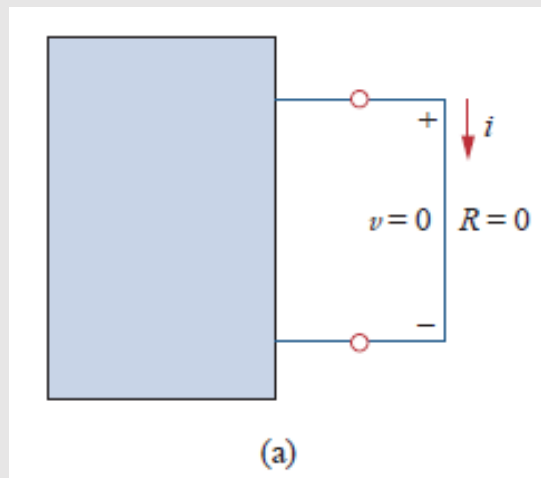
Short Circuit

An element with $R = 0$ is called a short circuit, as shown in Fig. (a). For a short circuit,

$$v = iR$$

showing that the voltage is zero but the current could be anything. In practice, a short circuit is usually a connecting wire assumed to be a perfect conductor. Thus,

A short circuit is a circuit element with resistance approaching zero



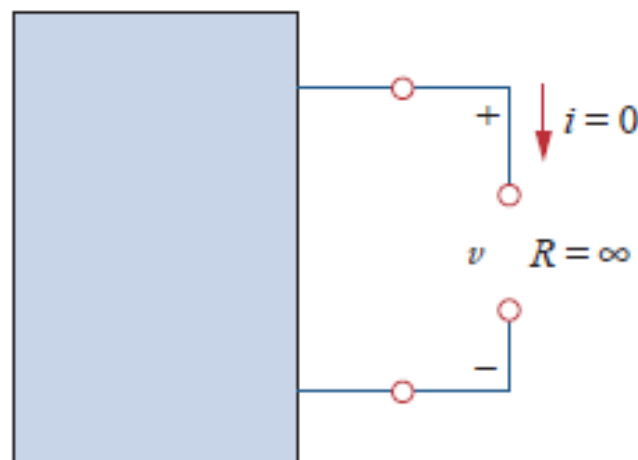
Open Circuit

Similarly, an element with $R = \infty$ is known as an *open circuit*, as shown in Fig. 2.2(b). For an open circuit,

$$i = \lim_{R \rightarrow \infty} \frac{v}{R} = 0 \quad (2.6)$$

indicating that the current is zero though the voltage could be anything. Thus,

An **open circuit** is a circuit element with resistance approaching infinity.



(b)

Conductance

A useful quantity in circuit analysis is the reciprocal of resistance R , known as *conductance* and denoted by G :

$$G = \frac{1}{R} = \frac{i}{v} \quad (2.7)$$

The conductance is a measure of how well an element will conduct electric current. The unit of conductance is the *mho* (ohm spelled backward) or reciprocal ohm, with symbol $\mathfrak{U}$, the inverted omega. Although engineers often use the mho, in this book we prefer to use the siemens (S), the SI unit of conductance:

$$1 \text{ S} = 1 \mathfrak{U} = 1 \text{ A/V} \quad (2.8)$$

Thus,

Conductance is the ability of an element to conduct electric current; it is measured in mhos ($\mathfrak{U}$) or siemens (S).

Power

The power dissipated by a resistor can be expressed in terms of R . Using Eqs. (1.7) and (2.3),

$$p = vi = i^2R = \frac{v^2}{R} \quad (2.10)$$

The power dissipated by a resistor may also be expressed in terms of G as

$$p = vi = v^2G = \frac{i^2}{G} \quad (2.11)$$

We should note two things from Eqs. (2.10) and (2.11):

1. The power dissipated in a resistor is a nonlinear function of either current or voltage.
2. Since R and G are positive quantities, the power dissipated in a resistor is always positive. Thus, a resistor always absorbs power from the circuit. This confirms the idea that a resistor is a passive element, incapable of generating energy.

Node, Branch and Loop

- ❑ Branch: A branch represents a single element such as a voltage source or a resistor.
- ❑ Node: A node is the point of connection between two or more branches.
- ❑ Loop: A loop is any closed path in a circuit.

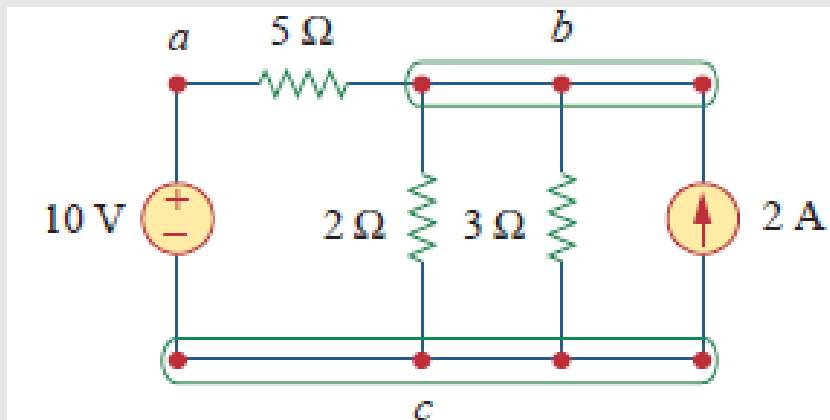


Figure 2.10

Nodes, branches, and loops.

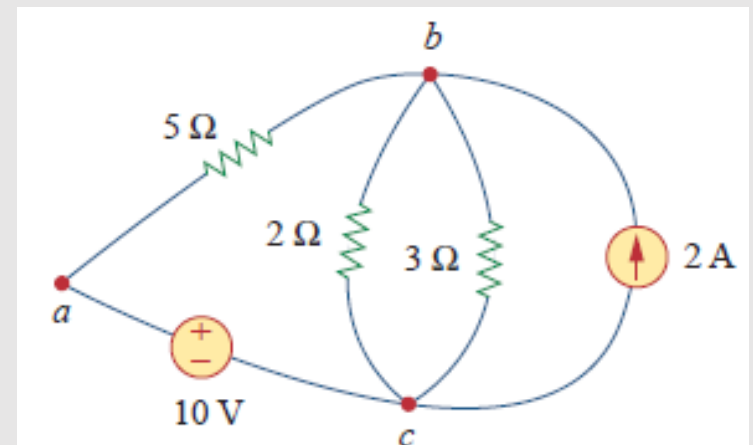


Figure 2.11

The three-node circuit of Fig. 2.10 is redrawn.

Kirchhoff's current law (KCL)

Kirchhoff's current law (KCL) states that the algebraic sum of currents entering a node (or a closed boundary) is zero.

Mathematically, KCL implies that

$$\sum_{n=1}^N i_n = 0$$

where N is the number of branches connected to the node and i_n is the n th current entering (or leaving) the node.

Kirchhoff's current law (KCL)

Consider the node in Fig. 2.16. Applying KCL gives

$$i_1 + (-i_2) + i_3 + i_4 + (-i_5) = 0 \quad (2.16)$$

since currents i_1 , i_3 , and i_4 are entering the node, while currents i_2 and i_5 are leaving it. By rearranging the terms, we get

$$i_1 + i_3 + i_4 = i_2 + i_5 \quad (2.17)$$

Equation (2.17) is an alternative form of KCL:

The sum of the currents entering a node is equal to the sum of the currents leaving the node.

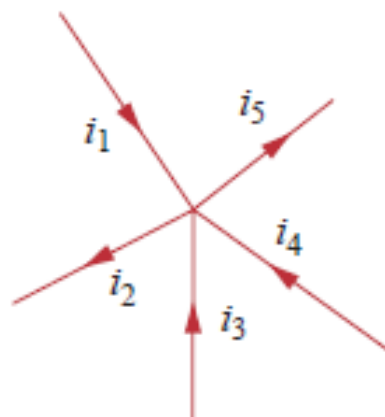


Figure 2.16

Currents at a node illustrating KCL.

Kirchhoff's voltage law (KVL)

Kirchhoff's voltage law (KVL) states that the algebraic sum of all voltages around a closed path (or loop) is zero.

Expressed mathematically, KVL states that

$$\sum_{m=1}^M v_m = 0$$

(2.19)

Thus, KVL yields

$$-v_1 + v_2 + v_3 - v_4 + v_5 = 0$$

Rearranging terms gives

$$v_2 + v_3 + v_5 = v_1 + v_4$$

which may be interpreted as

Sum of voltage drops = Sum of voltage rises

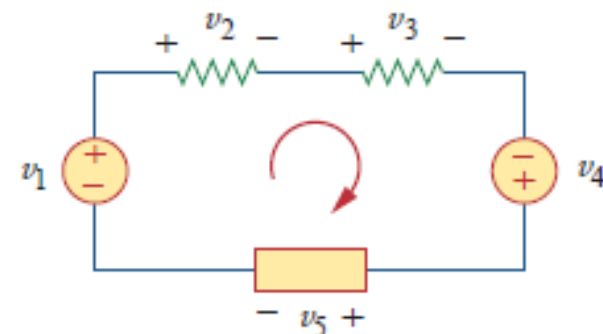


Figure 2.19

A single-loop circuit illustrating KVL.

Kirchhoff's voltage law (KVL)

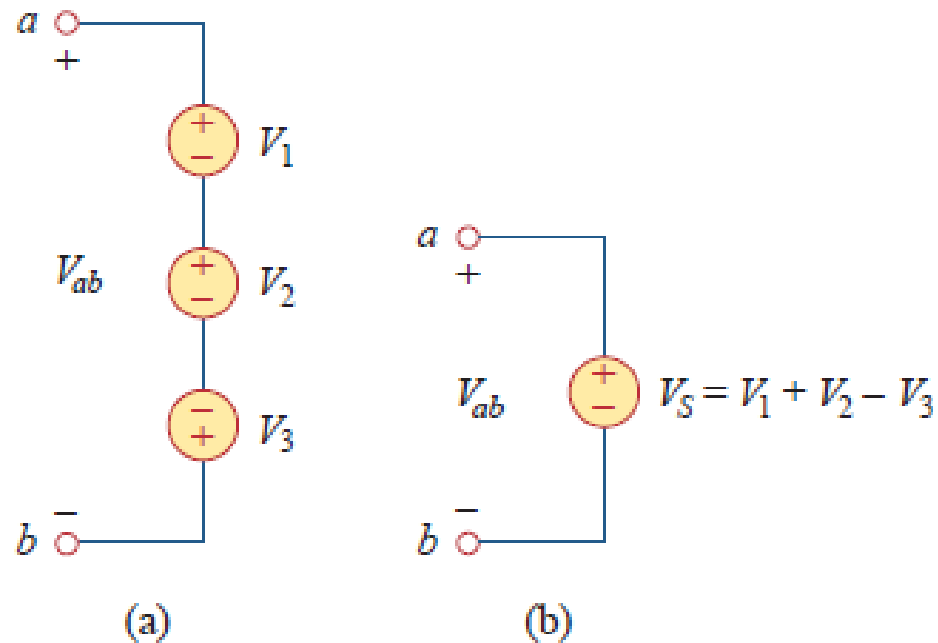


Figure 2.20

Voltage sources in series: (a) original circuit, (b) equivalent circuit.

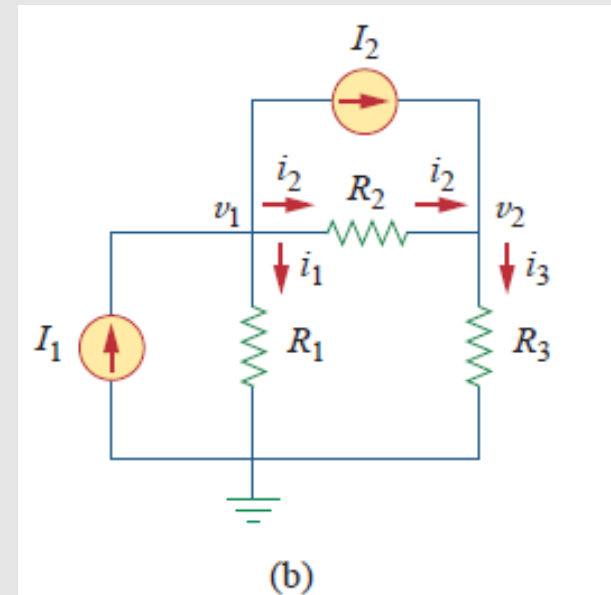
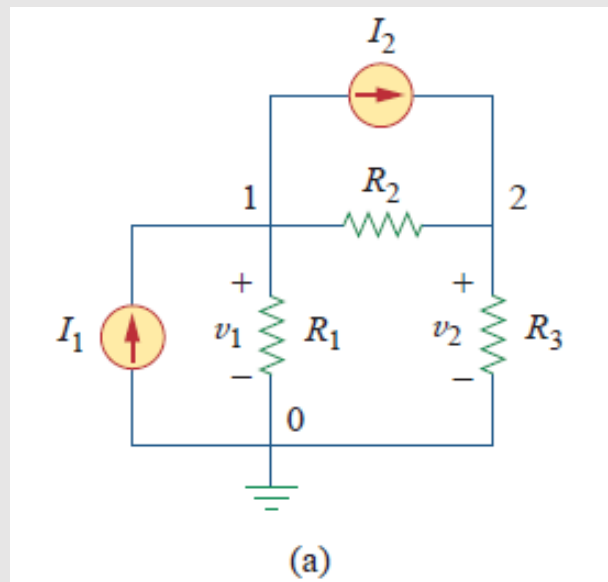
Nodal Analysis

- ❑ Nodal analysis provides a general procedure for analyzing circuits using node voltages as the circuit variables.

Steps to Determine Node Voltages:

1. Select a node as the reference node. Assign voltages $v_1, v_2, \dots, v_{n-1}$ to the remaining $n - 1$ nodes. The voltages are referenced with respect to the reference node.
2. Apply KCL to each of the $n - 1$ nonreference nodes. Use Ohm's law to express the branch currents in terms of node voltages.
3. Solve the resulting simultaneous equations to obtain the unknown node voltages.

Nodal Analysis



At node 1, applying KCL gives

$$I_1 = I_2 + i_1 + i_2$$

At node 2,

$$I_2 + i_2 = i_3$$

$$i = \frac{v_{\text{higher}} - v_{\text{lower}}}{R}$$

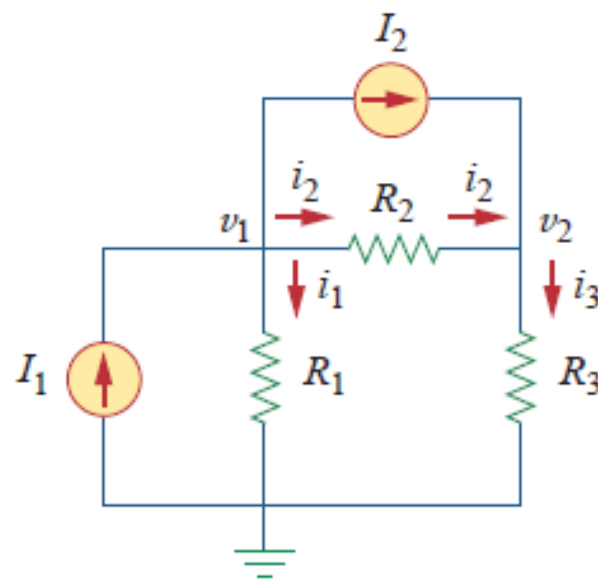
Nodal Analysis

Fig. 3.2(b),

$$\begin{aligned} i_1 &= \frac{v_1 - 0}{R_1} & \text{or} & & i_1 &= G_1 v_1 \\ i_2 &= \frac{v_1 - v_2}{R_2} & \text{or} & & i_2 &= G_2 (v_1 - v_2) \\ i_3 &= \frac{v_2 - 0}{R_3} & \text{or} & & i_3 &= G_3 v_2 \end{aligned} \quad (3.4)$$

Substituting Eq. (3.4) in Eqs. (3.1) and (3.2) results, respectively, in

$$\begin{aligned} I_1 &= I_2 + \frac{v_1}{R_1} + \frac{v_1 - v_2}{R_2} \\ I_2 + \frac{v_1 - v_2}{R_2} &= \frac{v_2}{R_3} \end{aligned}$$

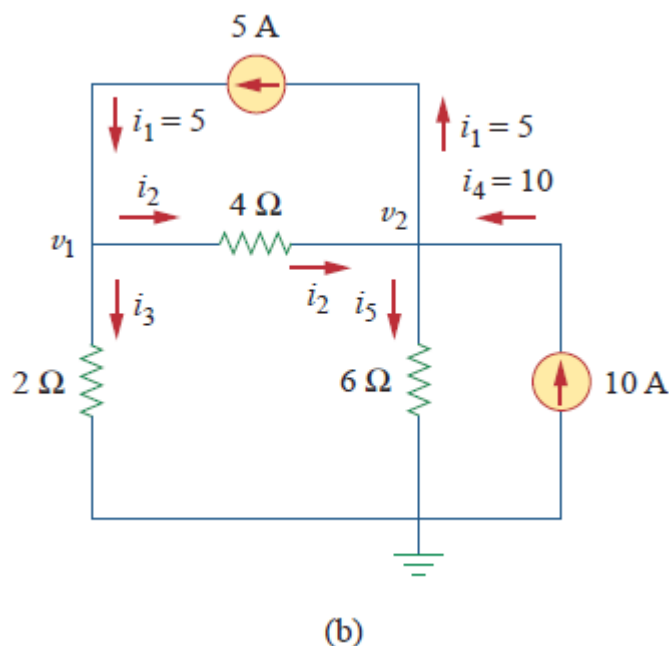
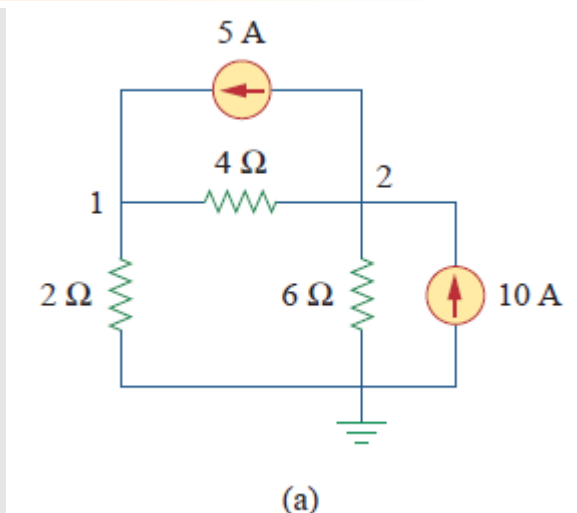


(b)

Nodal Analysis

Example 3.1

Calculate the node voltages in the circuit shown in Fig. 3.3(a).



At node 1, applying KCL and Ohm's law gives

$$i_1 = i_2 + i_3 \quad \Rightarrow \quad 5 = \frac{v_1 - v_2}{4} + \frac{v_1 - 0}{2}$$

Multiplying each term in the last equation by 4, we obtain

$$20 = v_1 - v_2 + 2v_1$$

or

$$3v_1 - v_2 = 20 \quad (3.1.1)$$

Nodal Analysis

At node 2, we do the same thing and get

$$i_2 + i_4 = i_1 + i_5 \quad \Rightarrow \quad \frac{v_1 - v_2}{4} + 10 = 5 + \frac{v_2 - 0}{6}$$

Multiplying each term by 12 results in

$$3v_1 - 3v_2 + 120 = 60 + 2v_2$$

or

$$-3v_1 + 5v_2 = 60 \quad (3.1.2)$$

Nodal Analysis: Equation Solve

■ **METHOD 1** Using the elimination technique, we add Eqs. (3.1.1) and (3.1.2).

$$4v_2 = 80 \quad \Rightarrow \quad v_2 = 20 \text{ V}$$

Substituting $v_2 = 20$ in Eq. (3.1.1) gives

$$3v_1 - 20 = 20 \quad \Rightarrow \quad v_1 = \frac{40}{3} = 13.333 \text{ V}$$

Nodal Analysis: Equation Solve

■ **METHOD 2** To use Cramer's rule, we need to put Eqs. (3.1.1 and (3.1.2) in matrix form as

$$\begin{bmatrix} 3 & -1 \\ -3 & 5 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 20 \\ 60 \end{bmatrix}$$

The determinant of the matrix is

$$\Delta = \begin{vmatrix} 3 & -1 \\ -3 & 5 \end{vmatrix} = 15 - 3 = 12$$

We now obtain v_1 and v_2 as

$$v_1 = \frac{\Delta_1}{\Delta} = \frac{\begin{vmatrix} 20 & -1 \\ 60 & 5 \end{vmatrix}}{\Delta} = \frac{100 + 60}{12} = 13.333 \text{ V}$$

$$v_2 = \frac{\Delta_2}{\Delta} = \frac{\begin{vmatrix} 3 & 20 \\ -3 & 60 \end{vmatrix}}{\Delta} = \frac{180 + 60}{12} = 20 \text{ V}$$

Example

Determine x_1 , x_2 , and x_3 for this set of simultaneous equations:

Example A.2

$$\begin{aligned} 25x_1 - 5x_2 - 20x_3 &= 50 \\ -5x_1 + 10x_2 - 4x_3 &= 0 \\ -5x_1 - 4x_2 + 9x_3 &= 0 \end{aligned}$$

Solution:

In matrix form, the given set of equations becomes

$$\begin{bmatrix} 25 & -5 & -20 \\ -5 & 10 & -4 \\ -5 & -4 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 50 \\ 0 \\ 0 \end{bmatrix}$$

We apply Eq. (A.11) to find the determinants. This requires that we repeat the first two rows of the matrix. Thus,

$$\begin{aligned} \Delta &= \begin{vmatrix} 25 & -5 & -20 \\ -5 & 10 & -4 \\ -5 & -4 & 9 \end{vmatrix} = \begin{vmatrix} 25 & -5 & -20 \\ -5 & 10 & -4 \\ -5 & -4 & 9 \\ 25 & -5 & -20 \\ -5 & 10 & -4 \end{vmatrix} \\ &= 25(10)9 + (-5)(-4)(-20) + (-5)(-5)(-4) \\ &\quad - (-20)(10)(-5) - (-4)(-4)25 - 9(-5)(-5) \\ &= 2250 - 400 - 100 - 1000 - 400 - 225 = 125 \end{aligned}$$

Similarly,

$$\begin{aligned} \Delta_1 &= \begin{vmatrix} 50 & -5 & -20 \\ 0 & 10 & -4 \\ 0 & -4 & 9 \end{vmatrix} = \begin{vmatrix} 50 & -5 & -20 \\ 0 & 10 & -4 \\ 0 & -4 & 9 \\ 50 & -5 & -20 \\ 0 & 10 & -4 \end{vmatrix} \\ &= 4500 + 0 + 0 - 0 - 800 - 0 = 3700 \end{aligned}$$

$$\Delta_2 = \begin{vmatrix} 25 & 50 & -20 \\ -5 & 0 & -4 \\ -5 & 0 & 9 \end{vmatrix} = \begin{vmatrix} 25 & 50 & -20 \\ -5 & 0 & -4 \\ -5 & 0 & 9 \end{vmatrix} =$$

$$= 0 + 0 + 1000 - 0 - 0 + 2250 = 3250$$

$$\Delta_3 = \begin{vmatrix} 25 & -5 & 50 \\ -5 & 10 & 0 \\ -5 & -4 & 0 \end{vmatrix} = \begin{vmatrix} 25 & -5 & 50 \\ -5 & 10 & 0 \\ -5 & -4 & 0 \end{vmatrix} =$$

$$= 0 + 1000 + 0 + 2500 - 0 - 0 = 3500$$

Hence, we now find

$$x_1 = \frac{\Delta_1}{\Delta} = \frac{3700}{125} = 29.6$$

$$x_2 = \frac{\Delta_2}{\Delta} = \frac{3250}{125} = 26$$

$$x_3 = \frac{\Delta_3}{\Delta} = \frac{3500}{125} = 28$$

Exercise Problem

Obtain the node voltages in the circuit of Fig. 3.4.

Answer: $v_1 = -2 \text{ V}$, $v_2 = -14 \text{ V}$.

Practice Problem 3.1

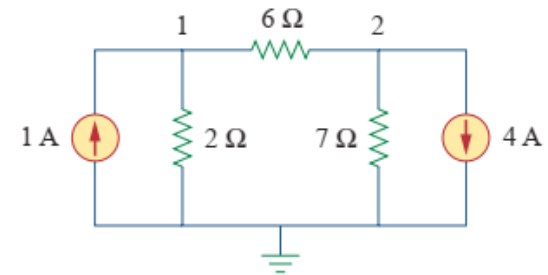


Figure 3.4
For Practice Prob. 3.1.

Example Problem

Determine the voltages at the nodes in Fig. 3.5(a).

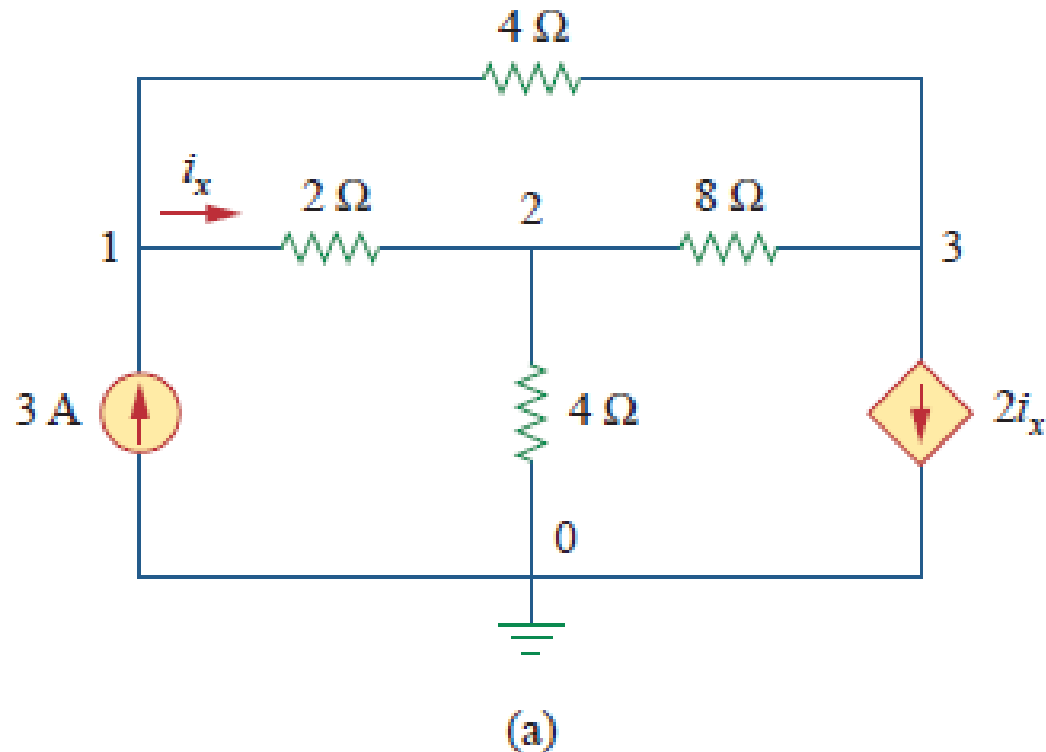
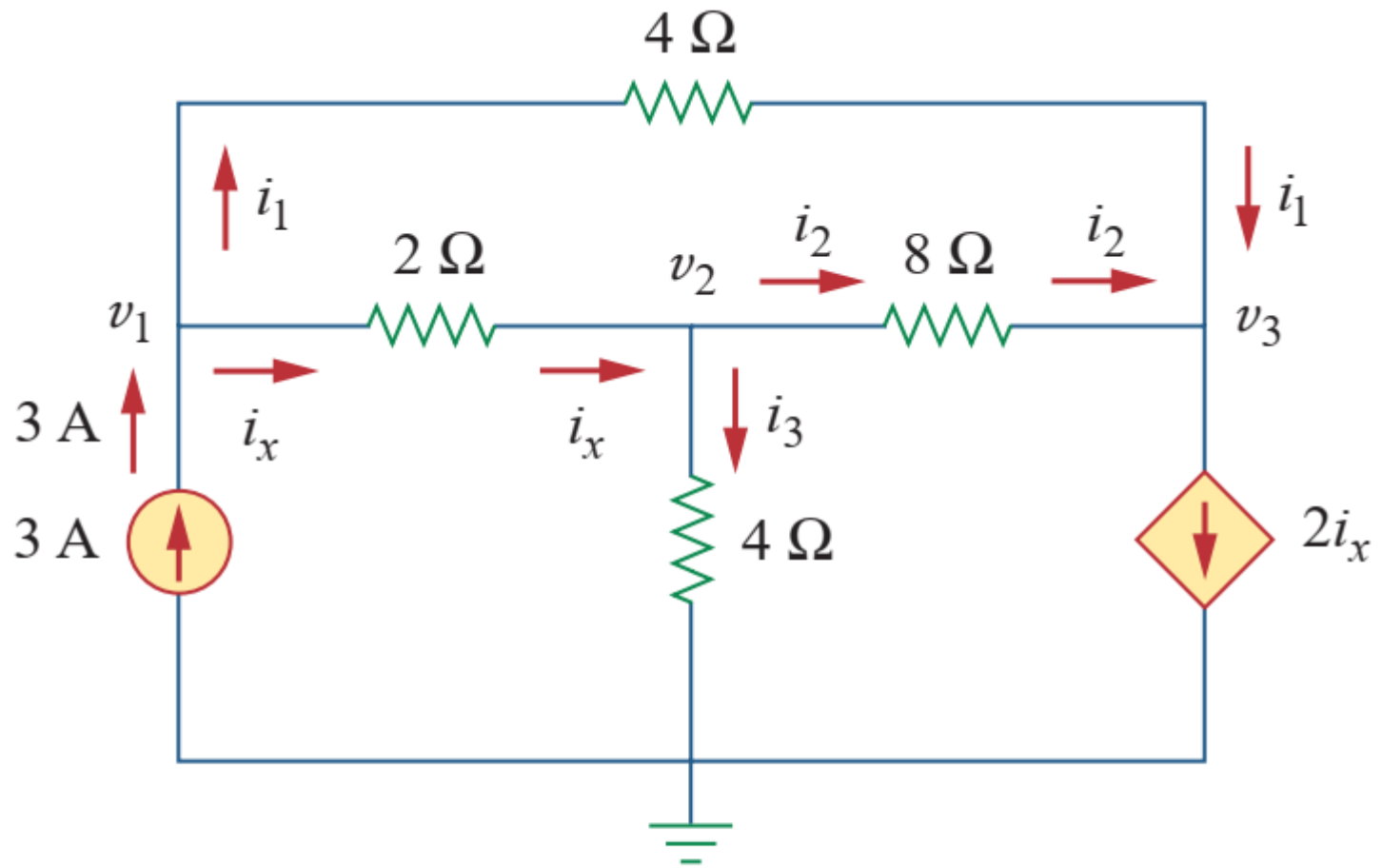


Figure 3.5

Example Problem



(b)

Example Problem

Figure 3.5

For Example 3.2: (a) original circuit, (b) circuit for analysis.

At node 1,

$$3 = i_1 + i_x \quad \Rightarrow \quad 3 = \frac{v_1 - v_3}{4} + \frac{v_1 - v_2}{2}$$

Multiplying by 4 and rearranging terms, we get

$$3v_1 - 2v_2 - v_3 = 12 \quad (3.2.1)$$

At node 2,

$$i_x = i_2 + i_3 \quad \Rightarrow \quad \frac{v_1 - v_2}{2} = \frac{v_2 - v_3}{8} + \frac{v_2 - 0}{4}$$

Multiplying by 8 and rearranging terms, we get

$$-4v_1 + 7v_2 - v_3 = 0 \quad (3.2.2)$$

At node 3,

$$i_1 + i_2 = 2i_x \quad \Rightarrow \quad \frac{v_1 - v_3}{4} + \frac{v_2 - v_3}{8} = \frac{2(v_1 - v_2)}{2}$$

Multiplying by 8, rearranging terms, and dividing by 3, we get

$$2v_1 - 3v_2 + v_3 = 0 \quad (3.2.3)$$

We have three simultaneous equations to solve to get the node voltages v_1 , v_2 , and v_3 . We shall solve the equations in three ways.

Thus,

$$v_1 = 4.8 \text{ V}, \quad v_2 = 2.4 \text{ V}, \quad v_3 = -2.4 \text{ V}$$

Exercise Problem

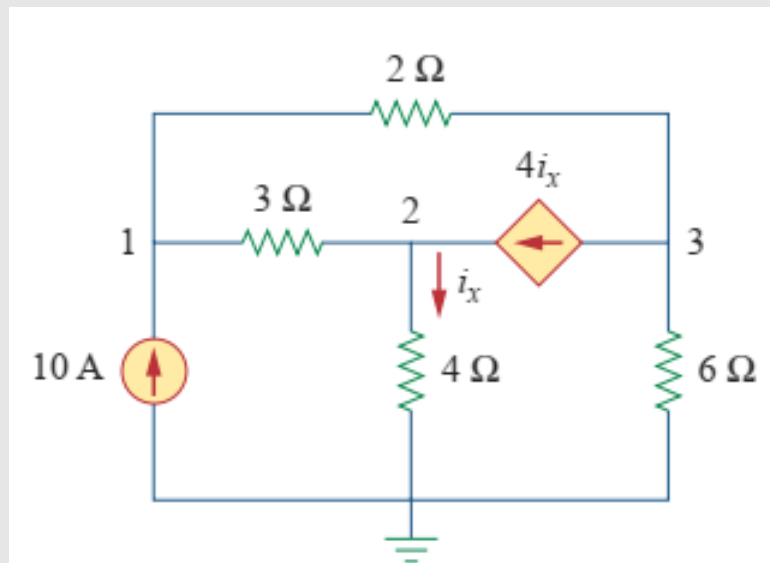


Figure 3.6
For Practice Prob. 3.2.

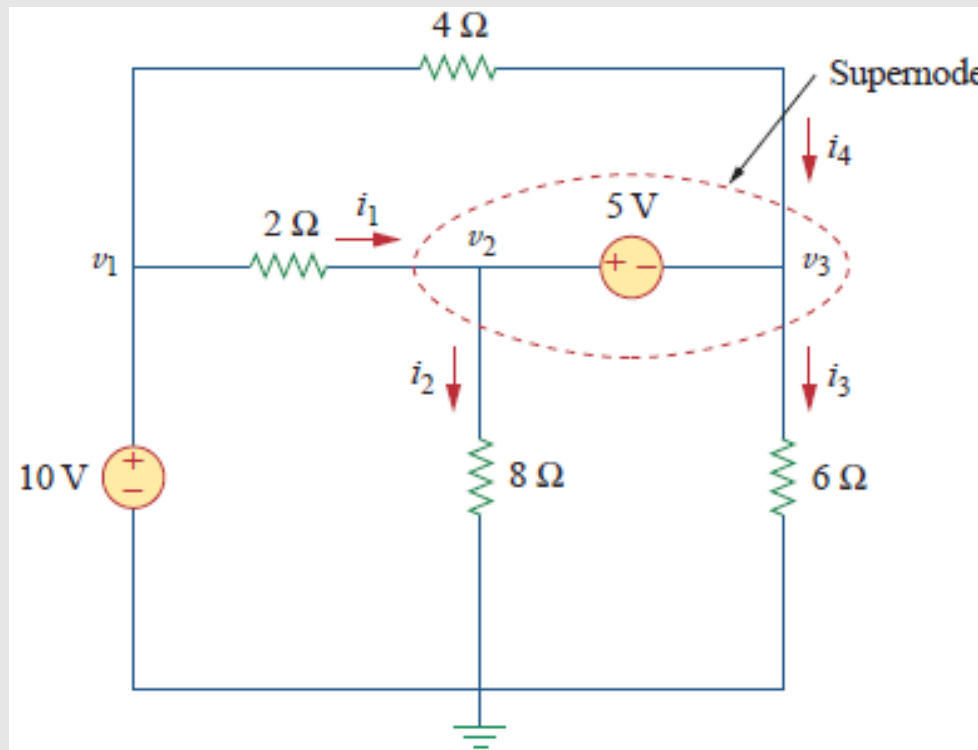
Find the voltages at the three nonreference nodes in the circuit of Fig. 3.6.

Answer: $v_1 = 80 \text{ V}$, $v_2 = -64 \text{ V}$, $v_3 = 156 \text{ V}$.

Nodal Analysis with Voltage Sources

- ❑ Case 1: If a voltage source is connected between the reference node and a nonreference node, we simply set the voltage at the nonreference node equal to the voltage of the voltage source.

In this following Figure, $v_1 = 10\text{ V}$



Nodal Analysis with Voltage Sources

- Case 2: If the voltage source (dependent or independent) is connected between two nonreference nodes, the two nonreference nodes form a *generalized node* or *supernode*; we apply both KCL and KVL to determine the node voltages.

A **supernode** is formed by enclosing a (dependent or independent) voltage source connected between two nonreference nodes and any elements connected in parallel with it.

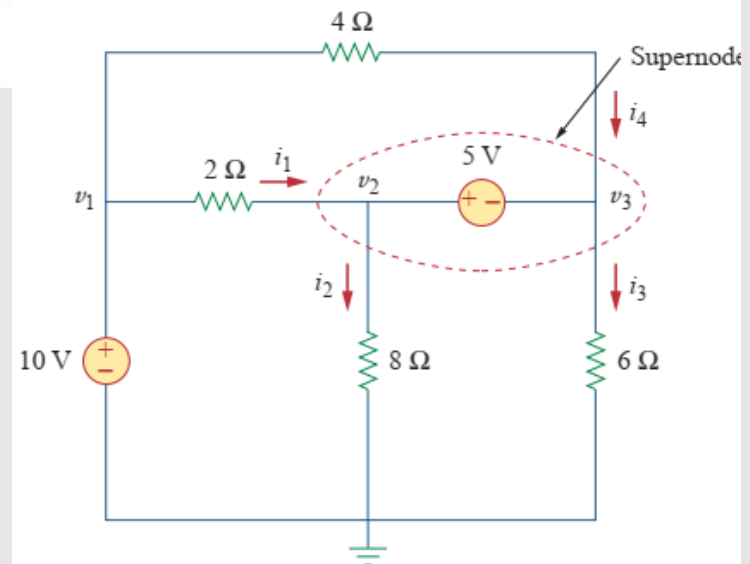


Figure 3.7

A circuit with a supernode.

Nodal Analysis: Case 2

Hence, at the super-

node in Fig. 3.7,

$$i_1 + i_4 = i_2 + i_3$$

or

$$\frac{v_1 - v_2}{2} + \frac{v_1 - v_3}{4} = \frac{v_2 - 0}{8} + \frac{v_3 - 0}{6}$$

To apply Kirchhoff's voltage law to the supernode in Fig. 3.7, we redraw the circuit as shown in Fig. 3.8. Going around the loop in the clockwise direction gives

$$-v_2 + 5 + v_3 = 0 \quad \Rightarrow \quad v_2 - v_3 = 5$$

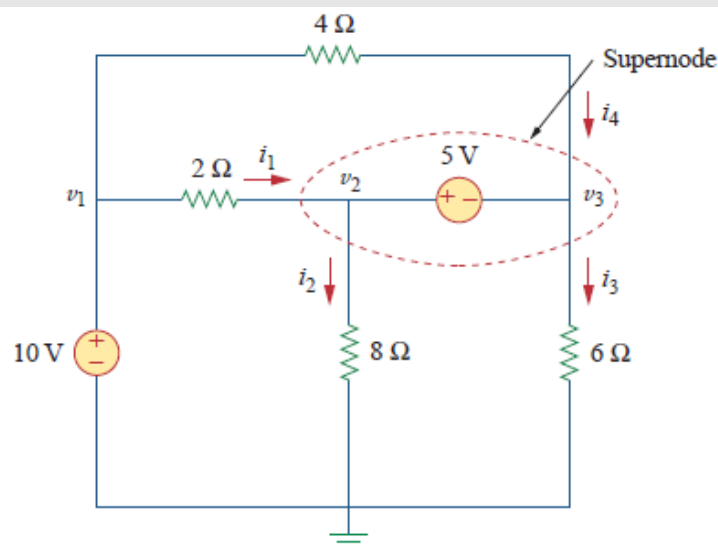


Figure 3.7

A circuit with a supernode.

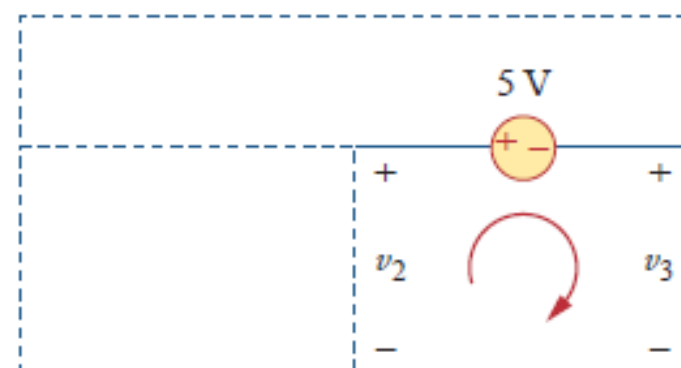


Figure 3.8

Applying KVL to a supernode.

Example Problem

For the circuit shown in Fig. 3.9, find the node voltages.

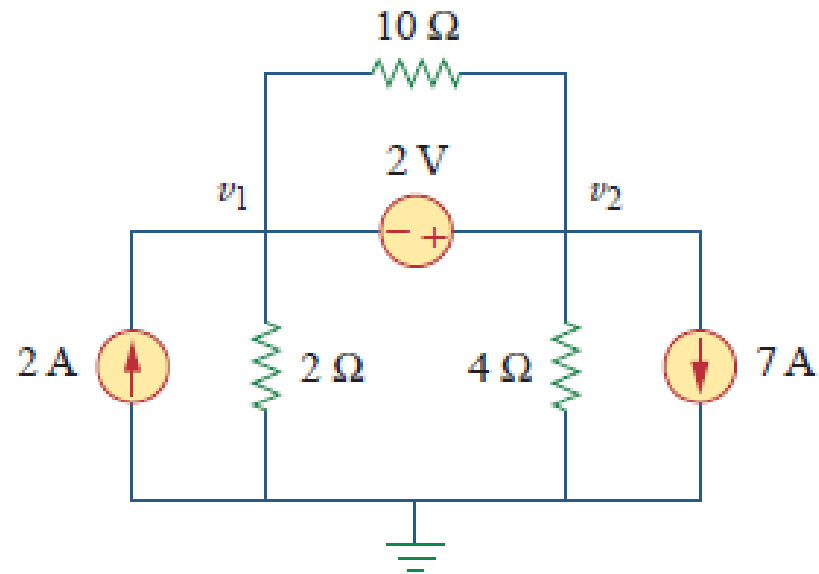


Figure 3.9
For Example 3.3.

Example Problem

Solution:

The supernode contains the 2-V source, nodes 1 and 2, and the 10- Ω resistor. Applying KCL to the supernode as shown in Fig. 3.10(a) gives

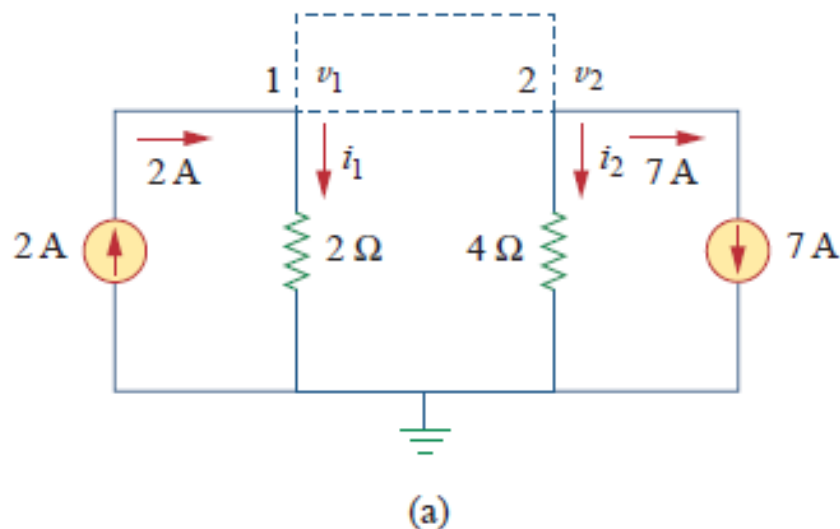
$$2 = i_1 + i_2 + 7$$

Expressing i_1 and i_2 in terms of the node voltages

$$2 = \frac{v_1 - 0}{2} + \frac{v_2 - 0}{4} + 7 \quad \Rightarrow \quad 8 = 2v_1 + v_2 + 28$$

or

$$v_2 = -20 - 2v_1 \quad (3.3.1)$$



Example Problem

To get the relationship between v_1 and v_2 , we apply KVL to the circuit in Fig. 3.10(b). Going around the loop, we obtain

$$-v_1 - 2 + v_2 = 0 \quad \Rightarrow \quad v_2 = v_1 + 2 \quad (3.3.2)$$

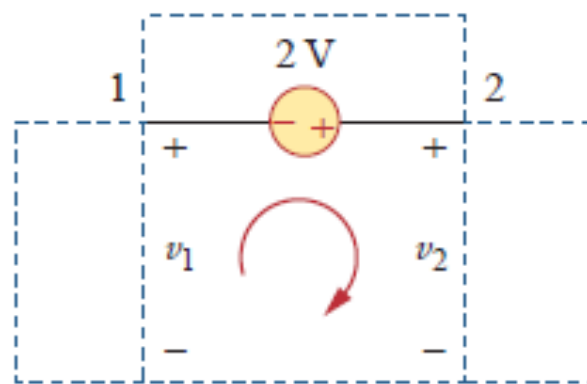
From Eqs. (3.3.1) and (3.3.2), we write

$$v_2 = v_1 + 2 = -20 - 2v_1$$

or

$$3v_1 = -22 \quad \Rightarrow \quad v_1 = -7.333 \text{ V}$$

and $v_2 = v_1 + 2 = -5.333 \text{ V}$. Note that the $10\text{-}\Omega$ resistor does not make any difference because it is connected across the supernode.



(b)

Exercise Problem

Practice Problem 3.3

Find v and i in the circuit of Fig. 3.11.

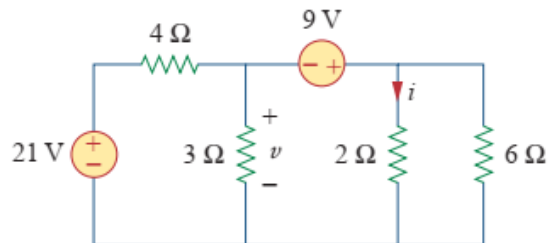


Figure 3.11

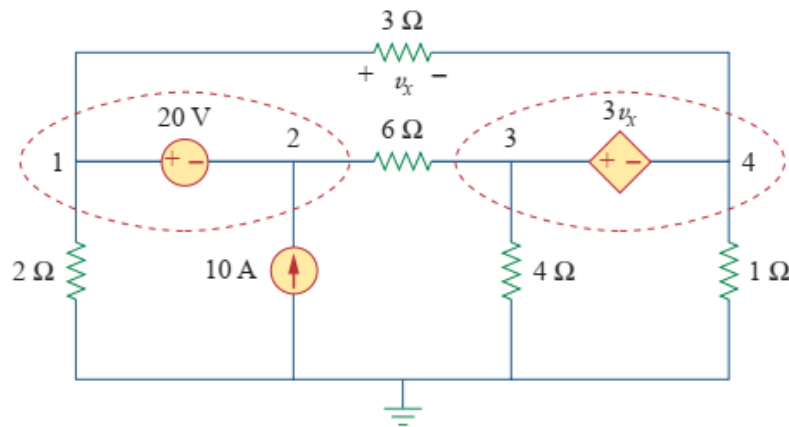
For Practice Prob. 3.3.

Answer: -0.6 V, 4.2 A.

Example Problem

Find the node voltages in the circuit of Fig. 3.12.

Example 3.4



$$V_{VL} = V_1 - V_4$$

Figure 3.12
For Example 3.4.

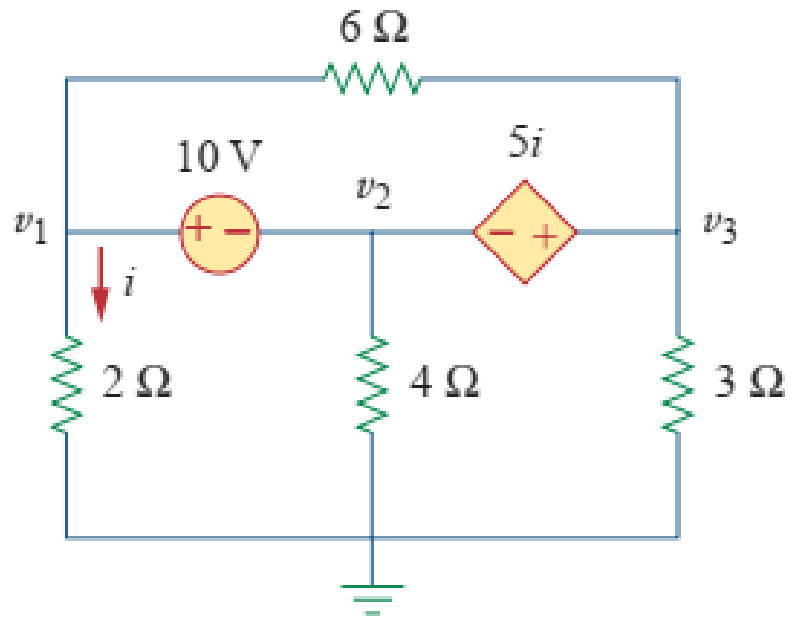
Thus, we arrive at the node voltages as

$$v_1 = \frac{\Delta_1}{\Delta} = \frac{-480}{-18} = 26.67 \text{ V}, \quad v_3 = \frac{\Delta_3}{\Delta} = \frac{-3120}{-18} = 173.33 \text{ V},$$

$$v_4 = \frac{\Delta_4}{\Delta} = \frac{840}{-18} = -46.67 \text{ V}$$

Exercise Problem

Practice Problem 3.4



Find v_1 , v_2 , and v_3 in the circuit of Fig. 3.14 using nodal analysis.

Answer: $v_1 = 3.043\text{ V}$, $v_2 = -6.956\text{ V}$, $v_3 = 0.6522\text{ V}$.

Mesh Analysis

- ❑ Mesh analysis provides another general procedure for analyzing circuits, using mesh currents as the circuit variables.
- ❑ A mesh is a loop which does not contain any other loops within it.
- ❑ Mesh analysis is only applicable to a circuit that is planar.
- ❑ A planar circuit is one that can be drawn in a plane with no branches crossing one another; otherwise it is nonplanar.

Mesh Analysis

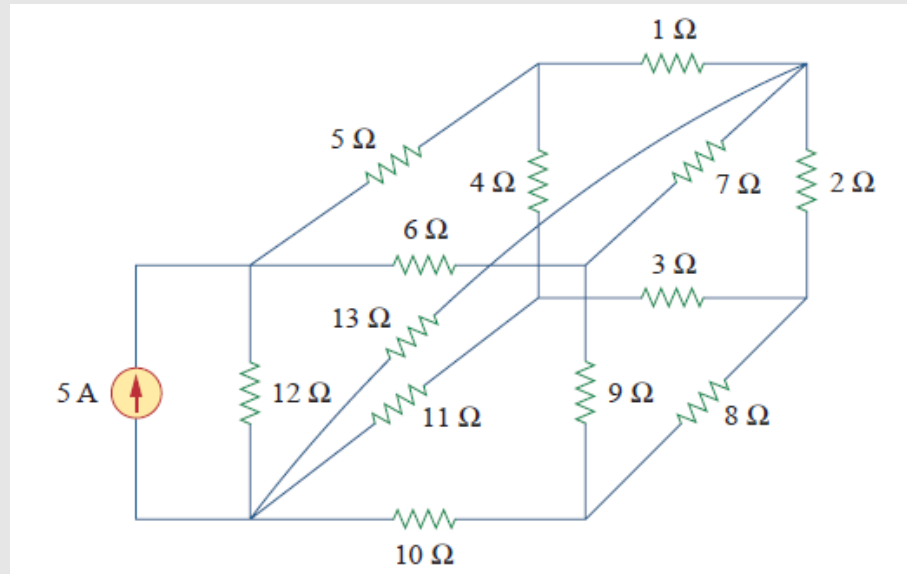


Figure: A nonplanar circuit.

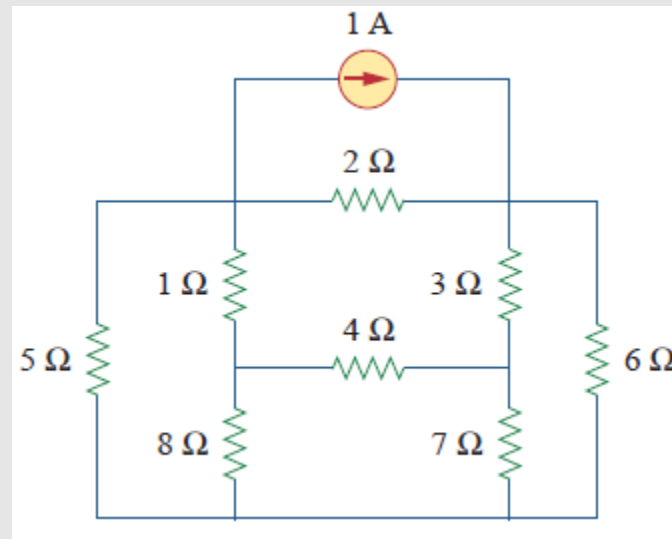


Figure: A planar circuit

Mesh Analysis

Steps to Determine Mesh Currents:

- ☐ Assign mesh currents to the n meshes.
- ☐ Apply KVL to each of the n meshes. Use Ohm's law to express the voltages in terms of the mesh currents.
- ☐ Solve the resulting n simultaneous equations to get the mesh currents.

Mesh Analysis

For mesh 1, we obtain

$$-V_1 + R_1 i_1 + R_3(i_1 - i_2) = 0$$

or

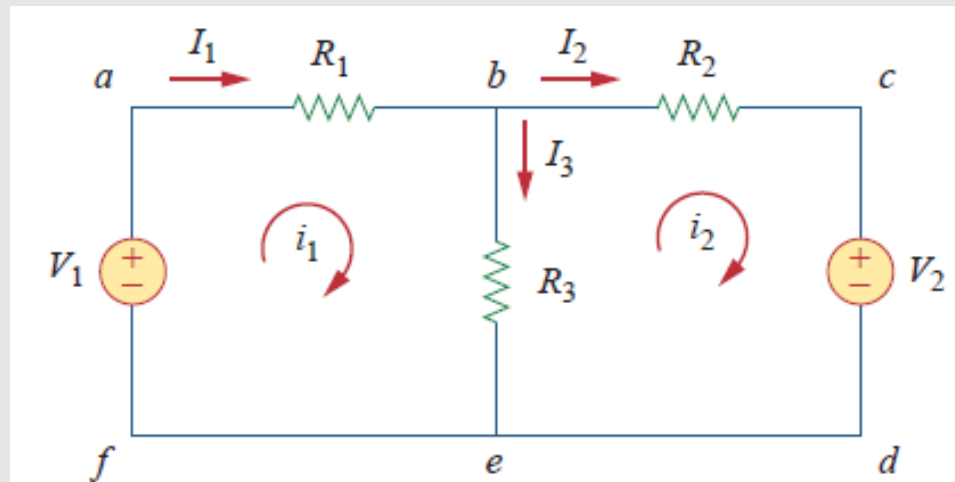
$$(R_1 + R_3)i_1 - R_3 i_2 = V_1$$

For mesh 2, applying KVL gives

$$R_2 i_2 + V_2 + R_3(i_2 - i_1) = 0$$

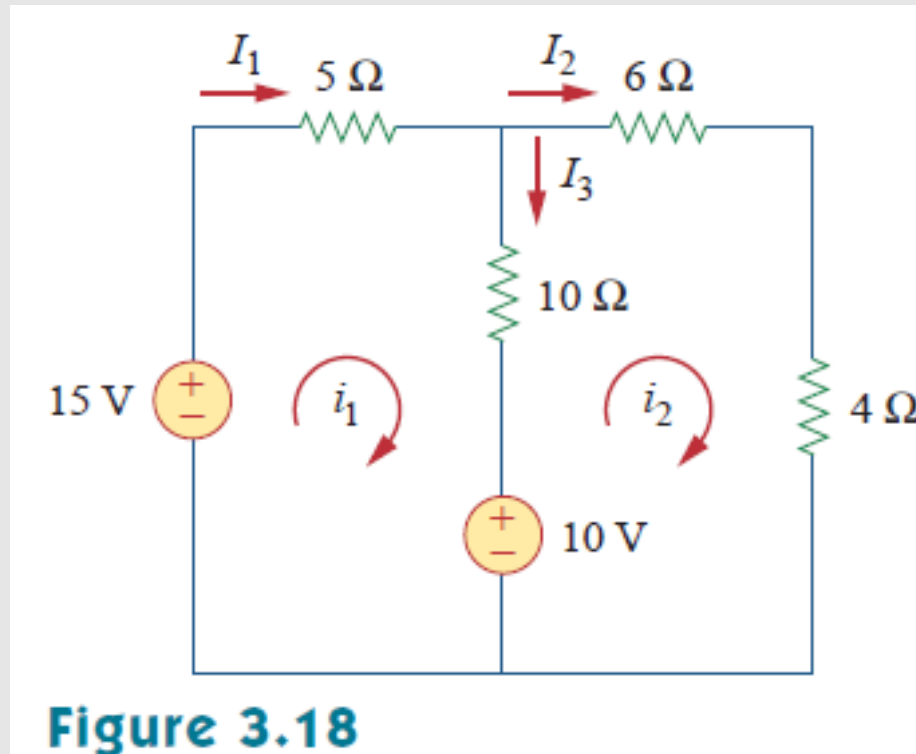
or

$$-R_3 i_1 + (R_2 + R_3)i_2 = -V_2$$



Mesh Analysis: Example Problem

For the circuit in Fig. 3.18, find the branch currents and using mesh analysis.



Mesh Analysis: Example Problem

Solution:

We first obtain the mesh currents using KVL. For mesh 1,

$$-15 + 5i_1 + 10(i_1 - i_2) + 10 = 0$$

or

$$3i_1 - 2i_2 = 1$$

For mesh 2,

$$6i_2 + 4i_2 + 10(i_2 - i_1) - 10 = 0$$

or

$$i_1 = 2i_2 - 1$$

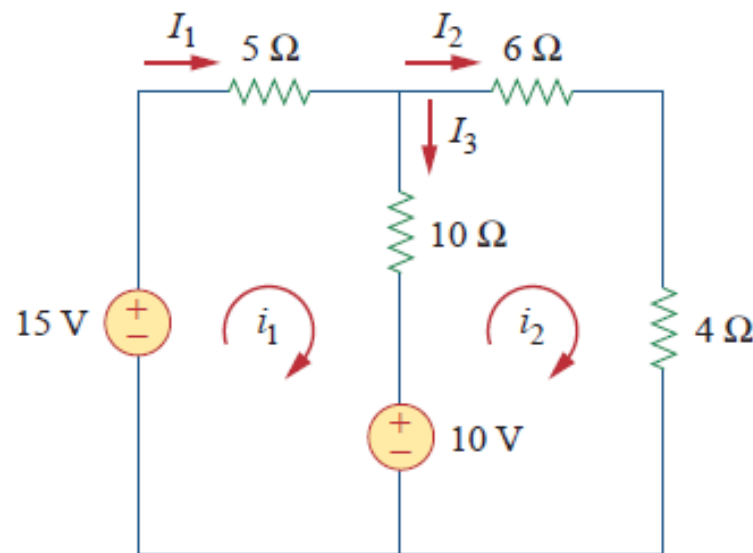


Figure 3.18

Mesh Analysis: Example Problem

■ **METHOD 2** To use Cramer's rule, we cast Eqs. (3.5.1) and (3.5.2) in matrix form as

$$\begin{bmatrix} 3 & -2 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

We obtain the determinants

$$\Delta = \begin{vmatrix} 3 & -2 \\ -1 & 2 \end{vmatrix} = 6 - 2 = 4$$

$$\Delta_1 = \begin{vmatrix} 1 & -2 \\ 1 & 2 \end{vmatrix} = 2 + 2 = 4, \quad \Delta_2 = \begin{vmatrix} 3 & 1 \\ -1 & 1 \end{vmatrix} = 3 + 1 = 4$$

Thus,

$$i_1 = \frac{\Delta_1}{\Delta} = 1 \text{ A}, \quad i_2 = \frac{\Delta_2}{\Delta} = 1 \text{ A}$$

as before.

Exercise Problem

Practice Problem 3.5

Calculate the mesh currents i_1 and i_2 of the circuit of Fig. 3.19.

Answer: $i_1 = 2 \text{ A}$, $i_2 = 0 \text{ A}$.

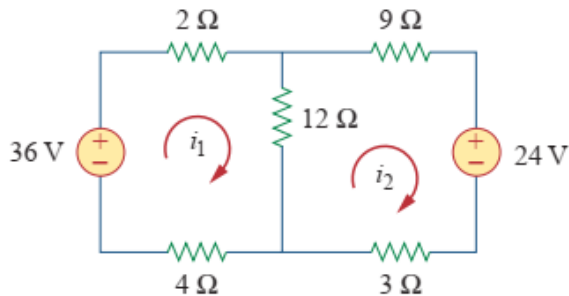
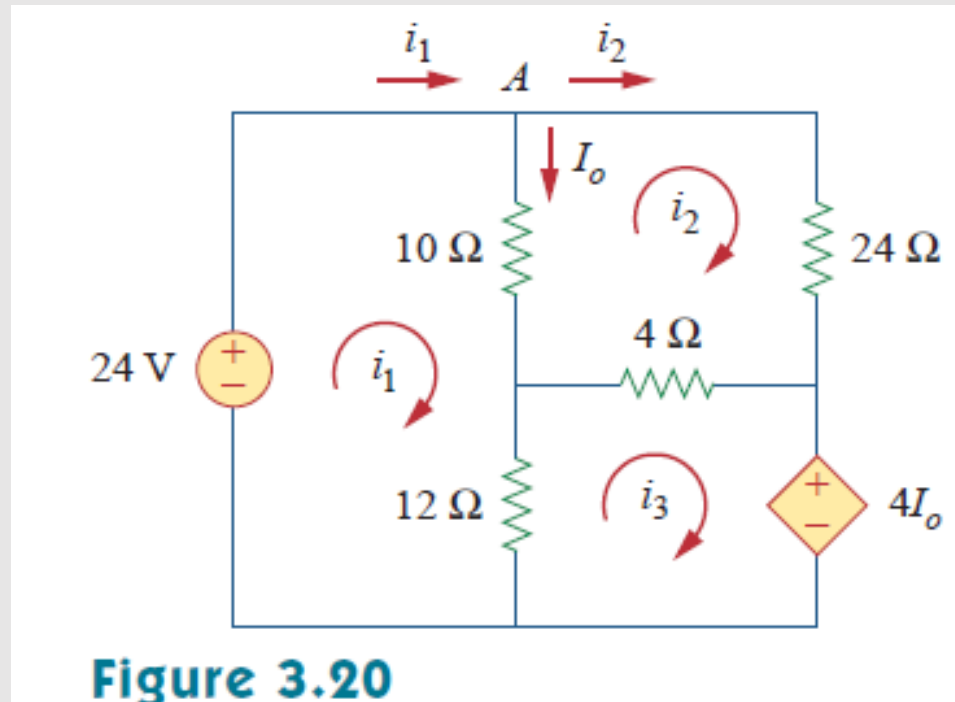


Figure 3.19

For Practice Prob. 3.5.

Mesh Analysis: Example Problem

Use mesh analysis to find the current in the circuit of Fig. 3.20.



Mesh Analysis: Example Problem

Solution:

We apply KVL to the three meshes in turn. For mesh 1,

$$-24 + 10(i_1 - i_2) + 12(i_1 - i_3) = 0$$

or

$$11i_1 - 5i_2 - 6i_3 = 12$$

For mesh 2,

$$24i_2 + 4(i_2 - i_3) + 10(i_2 - i_1) = 0$$

or

$$-5i_1 + 19i_2 - 2i_3 = 0$$

Mesh Analysis: Example Problem

For mesh 3,

$$4I_o + 12(i_3 - i_1) + 4(i_3 - i_2) = 0$$

But at node A, $I_o = i_1 - i_2$, so that

$$4(i_1 - i_2) + 12(i_3 - i_1) + 4(i_3 - i_2) = 0$$

or

$$-i_1 - i_2 + 2i_3 = 0$$

In matrix form,

$$\begin{bmatrix} 11 & -5 & -6 \\ -5 & 19 & -2 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \\ 0 \end{bmatrix}$$

Mesh Analysis: Example Problem

$$i_1 = \frac{\Delta_1}{\Delta} = \frac{432}{192} = 2.25 \text{ A}, \quad i_2 = \frac{\Delta_2}{\Delta} = \frac{144}{192} = 0.75 \text{ A},$$

$$i_3 = \frac{\Delta_3}{\Delta} = \frac{288}{192} = 1.5 \text{ A}$$

Thus, $I_o = i_1 - i_2 = 1.5 \text{ A}$.

Exercise Problem

Practice Problem 3.6

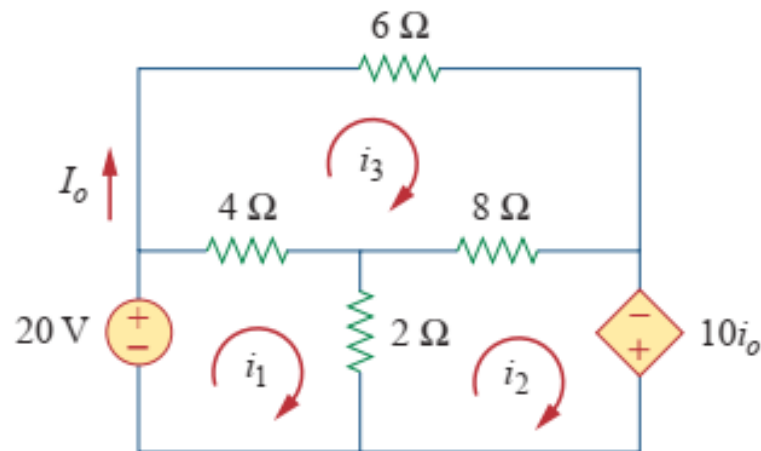


Figure 3.21

For Practice Prob. 3.6.

Using mesh analysis, find I_o in the circuit of Fig. 3.21.

Answer: -5 A.

Mesh Analysis with Current Sources

■ **CASE 1** When a current source exists only in one mesh: Consider the circuit in Fig. 3.22, for example. We set $i_2 = -5$ A and write a mesh equation for the other mesh in the usual way; that is,

$$-10 + 4i_1 + 6(i_1 - i_2) = 0 \quad \Rightarrow \quad i_1 = -2 \text{ A}$$

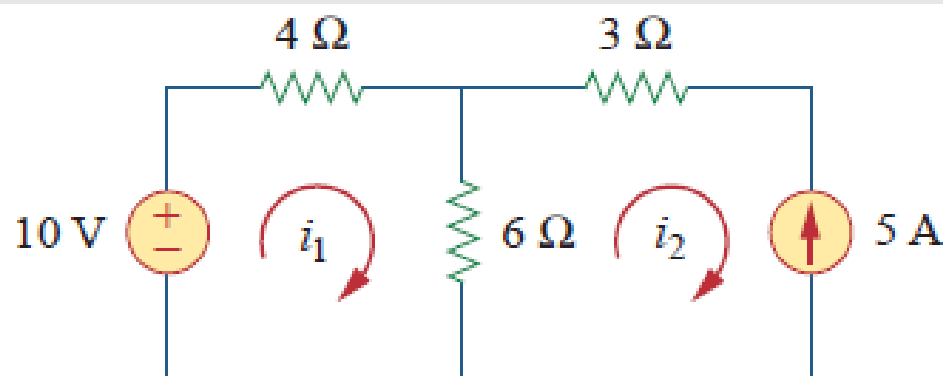


Figure 3.22

A circuit with a current source.

Mesh Analysis with Current Sources

■ **CASE 2** When a current source exists between two meshes: Consider the circuit in Fig. 3.23(a), for example. We create a *supermesh* by excluding the current source and any elements connected in series with it, as shown in Fig. 3.23(b). Thus,

A **supermesh** results when two meshes have a (dependent or independent) current source in common.

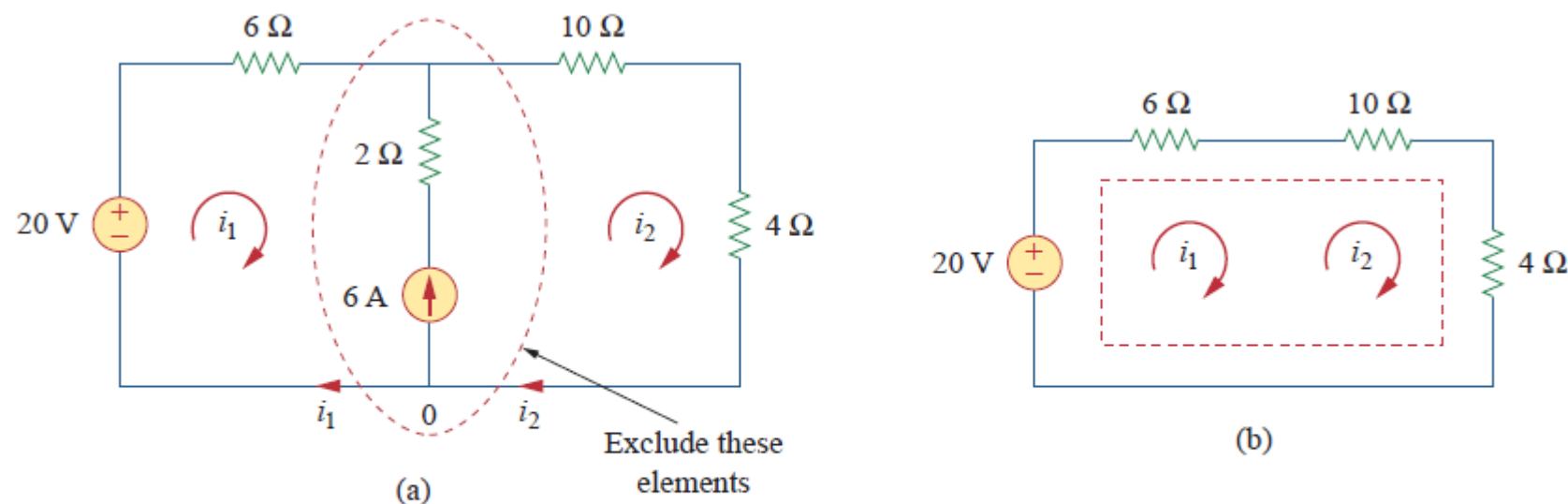


Figure 3.23

(a) Two meshes having a current source in common, (b) a supermesh, created by excluding the current source.

Mesh Analysis with Current Sources

For the circuit in Fig. 3.24, find i_1 to i_4 using mesh analysis.

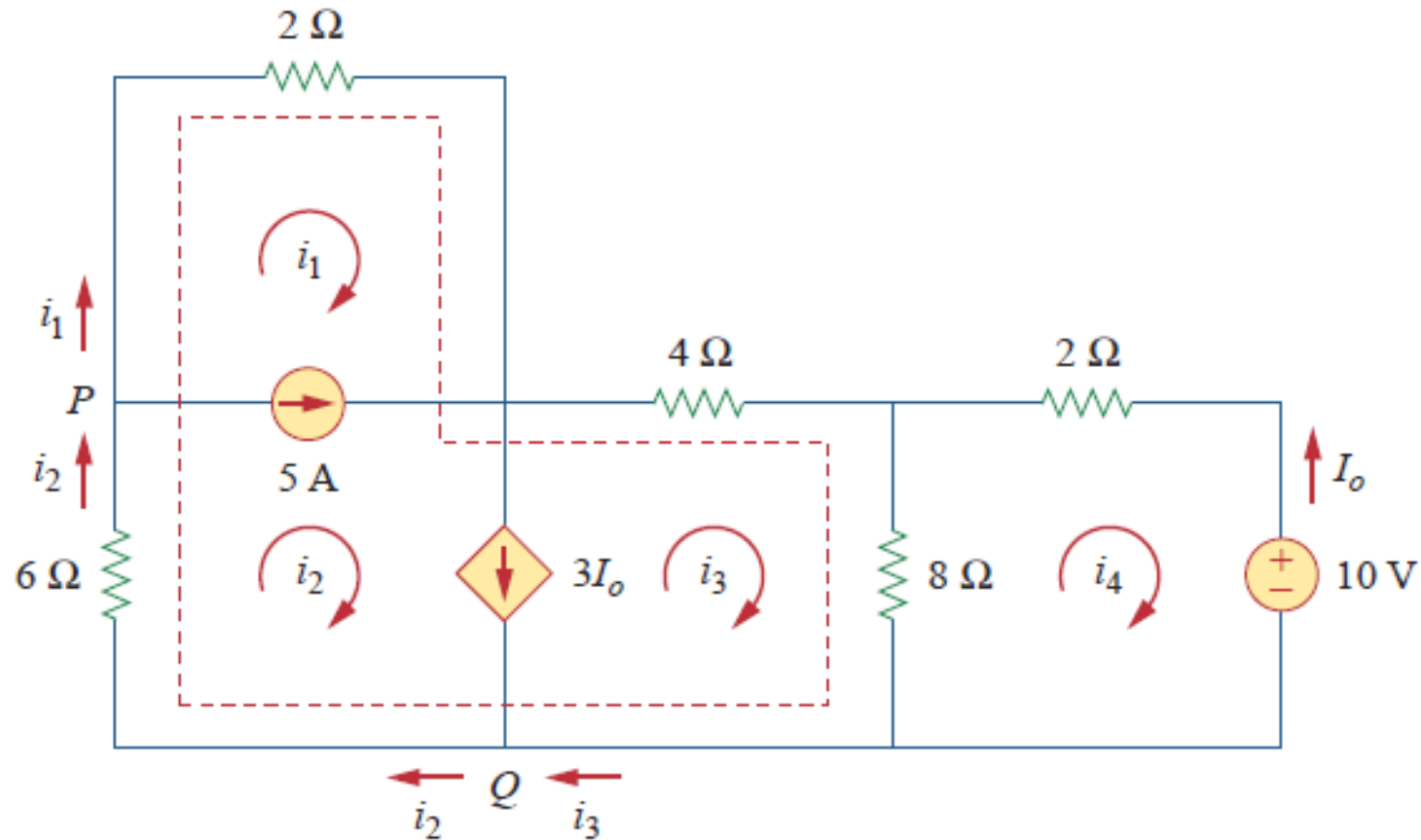


Figure 3.24

For Example 3.7.

Mesh Analysis with Current Sources

Solution:

Note that meshes 1 and 2 form a supermesh since they have an independent current source in common. Also, meshes 2 and 3 form another supermesh because they have a dependent current source in common. The two supermeshes intersect and form a larger supermesh as shown. Applying KVL to the larger supermesh,

$$2i_1 + 4i_3 + 8(i_3 - i_4) + 6i_2 = 0$$

or

$$i_1 + 3i_2 + 6i_3 - 4i_4 = 0 \quad (3.7.1)$$

For the independent current source, we apply KCL to node P :

$$i_2 = i_1 + 5 \quad (3.7.2)$$

For the dependent current source, we apply KCL to node Q :

$$i_2 = i_3 + 3I_o$$

Mesh Analysis with Current Sources

But $I_o = -i_4$, hence,

$$i_2 = i_3 - 3i_4 \quad (3.7.3)$$

Applying KVL in mesh 4,

$$2i_4 + 8(i_4 - i_3) + 10 = 0$$

or

$$5i_4 - 4i_3 = -5 \quad (3.7.4)$$

From Eqs. (3.7.1) to (3.7.4),

$$i_1 = -7.5 \text{ A}, \quad i_2 = -2.5 \text{ A}, \quad i_3 = 3.93 \text{ A}, \quad i_4 = 2.143 \text{ A}$$

Exercise Problem

Practice Problem 3.7

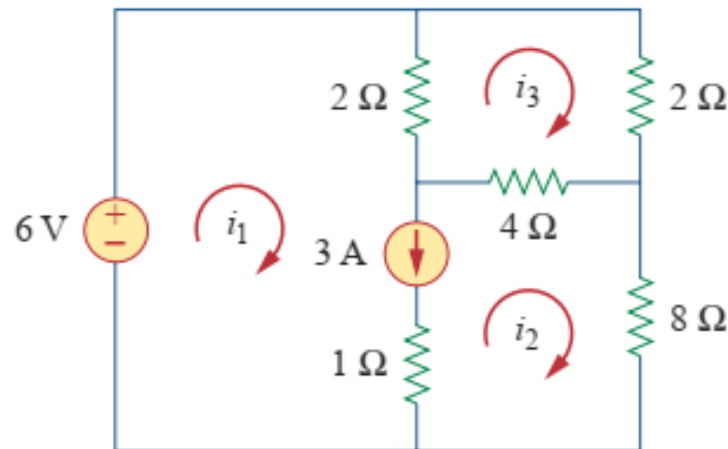


Figure 3.25

For Practice Prob. 3.7.

Use mesh analysis to determine i_1 , i_2 , and i_3 in Fig. 3.25.

Answer: $i_1 = 3.474$ A, $i_2 = 0.4737$ A, $i_3 = 1.1052$ A.

Circuit Theorems

involving Dependent Source

Superposition

- ❑ The superposition principle states that the voltage across (or current through) an element in a linear circuit is the algebraic sum of the voltages across (or currents through) that element due to each independent source acting alone.

- ❑ Steps to apply Superposition Principle:
 - a) Turn off all independent sources except one source. Find the output (voltage or current) due to that active source using any method like nodal analysis/mesh analysis/KVL/KCL.
 - b) Repeat step (a) for each of the other independent sources.
 - c) Find the total contribution by adding algebraically all the contributions due to the independent sources.

Superposition

Example 4.4

Find i_o in the circuit of Fig. 4.9 using superposition.

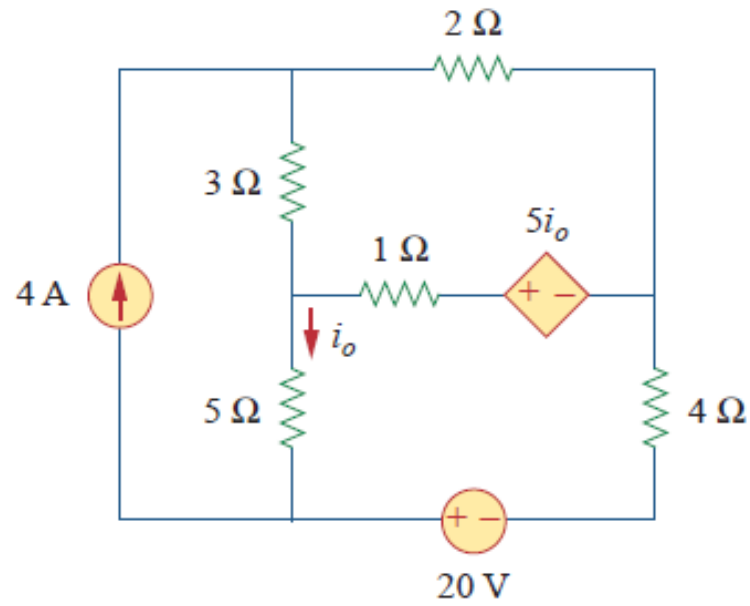


Figure 4.9

For Example 4.4.

Superposition

Solution:

The circuit in Fig. 4.9 involves a dependent source, which must be left intact. We let

$$i_o = i'_o + i''_o \quad (4.4.1)$$

where i'_o and i''_o are due to the 4-A current source and 20-V voltage source respectively.

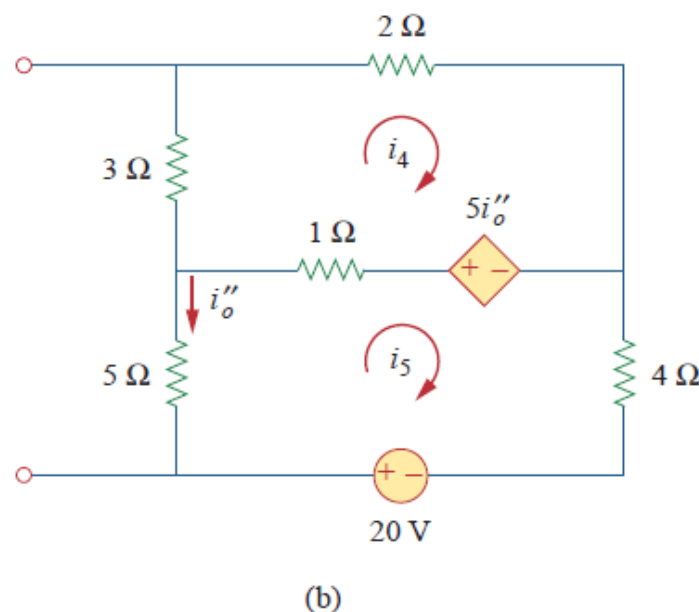
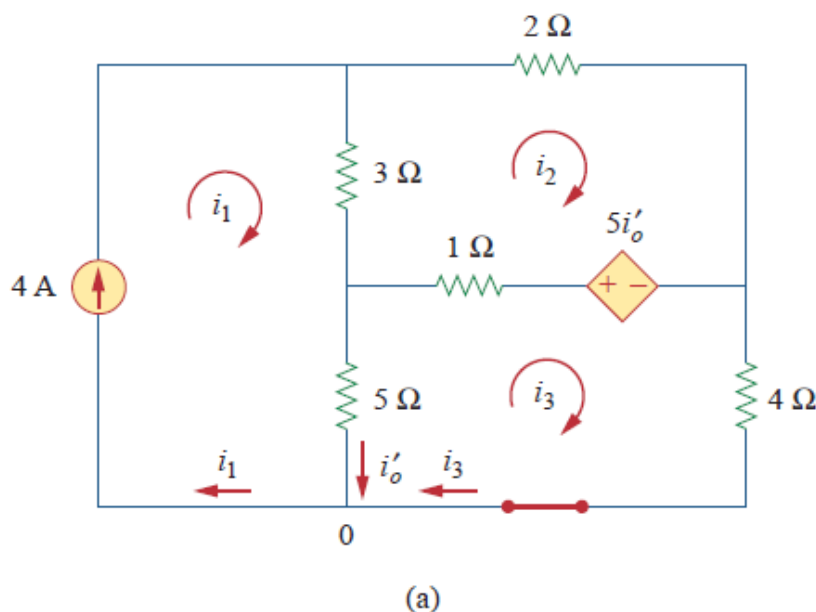


Figure 4.10

For Example 4.4: Applying superposition to (a) obtain i'_o , (b) obtain i''_o .

Superposition

To obtain i'_o , we turn off the 20-V source so that we have the circuit in Fig. 4.10(a). We apply mesh analysis in order to obtain i'_o . For loop 1,

$$i_1 = 4 \text{ A} \quad (4.4.2)$$

For loop 2,

$$-3i_1 + 6i_2 - 1i_3 - 5i'_o = 0 \quad (4.4.3)$$

For loop 3,

$$-5i_1 - 1i_2 + 10i_3 + 5i'_o = 0 \quad (4.4.4)$$

But at node 0,

$$i_3 = i_1 - i'_o = 4 - i'_o \quad (4.4.5)$$

Substituting Eqs. (4.4.2) and (4.4.5) into Eqs. (4.4.3) and (4.4.4) gives two simultaneous equations

$$3i_2 - 2i'_o = 8 \quad (4.4.6)$$

$$i_2 + 5i'_o = 20 \quad (4.4.7)$$

which can be solved to get

$$i'_o = \frac{52}{17} \text{ A} \quad (4.4.8)$$

Superposition

To obtain i_o'' , we turn off the 4-A current source so that the circuit becomes that shown in Fig. 4.10(b). For loop 4, KVL gives

$$6i_4 - i_5 - 5i_o'' = 0 \quad (4.4.9)$$

and for loop 5,

$$-i_4 + 10i_5 - 20 + 5i_o'' = 0 \quad (4.4.10)$$

But $i_5 = -i_o''$. Substituting this in Eqs. (4.4.9) and (4.4.10) gives

$$6i_4 - 4i_o'' = 0 \quad (4.4.11)$$

$$i_4 + 5i_o'' = -20 \quad (4.4.12)$$

which we solve to get

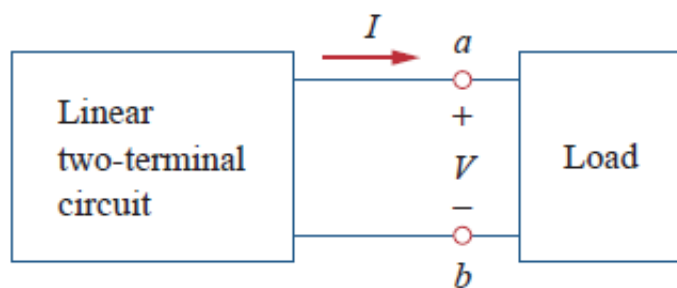
$$i_o'' = -\frac{60}{17} \text{ A} \quad (4.4.13)$$

Now substituting Eqs. (4.4.8) and (4.4.13) into Eq. (4.4.1) gives

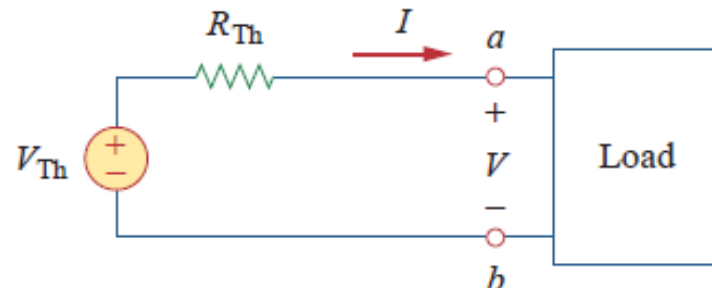
$$i_o = -\frac{8}{17} = -0.4706 \text{ A}$$

Thevenin's Theorem

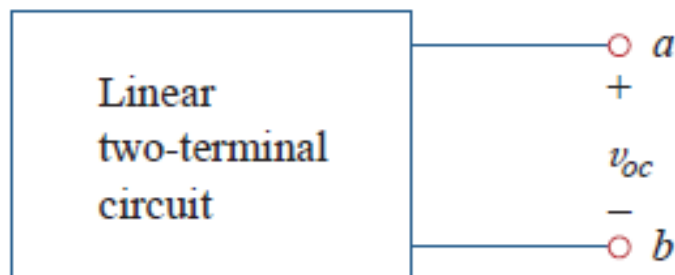
Thevenin's theorem states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a voltage source V_{Th} in series with a resistor R_{Th} , where V_{Th} is the open-circuit voltage at the terminals and R_{Th} is the input or equivalent resistance at the terminals when the independent sources are turned off.



(a)

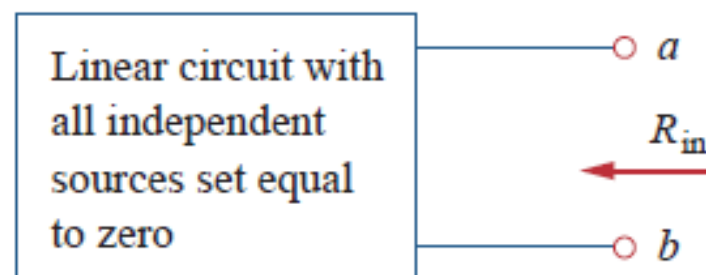


(b)



$$V_{Th} = v_{oc}$$

(a)



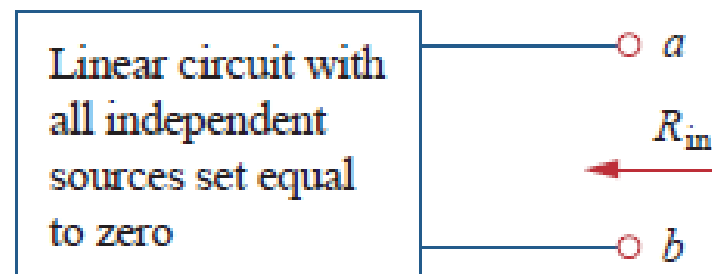
$$R_{Th} = R_{in}$$

(b)

Thevenin's Theorem

To apply this idea in finding the Thevenin resistance R_{Th} , we need to consider two cases.

■ **CASE 1** If the network has no dependent sources, we turn off all independent sources. R_{Th} is the input resistance of the network looking between terminals a and b , as shown in Fig. 4.24(b).

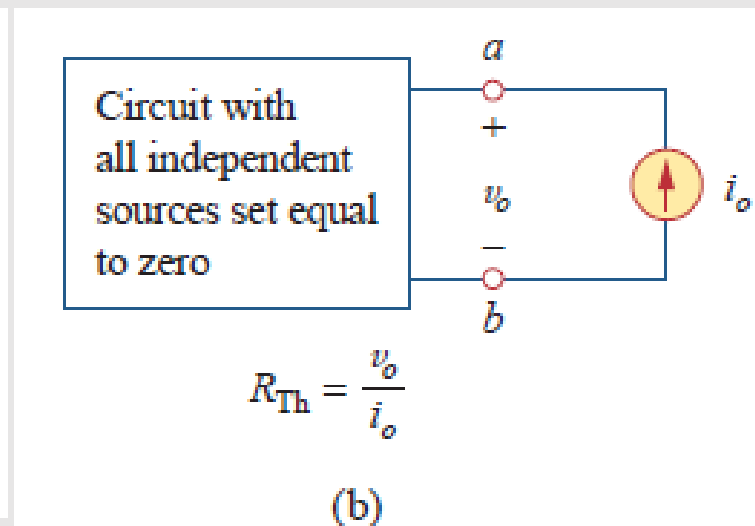
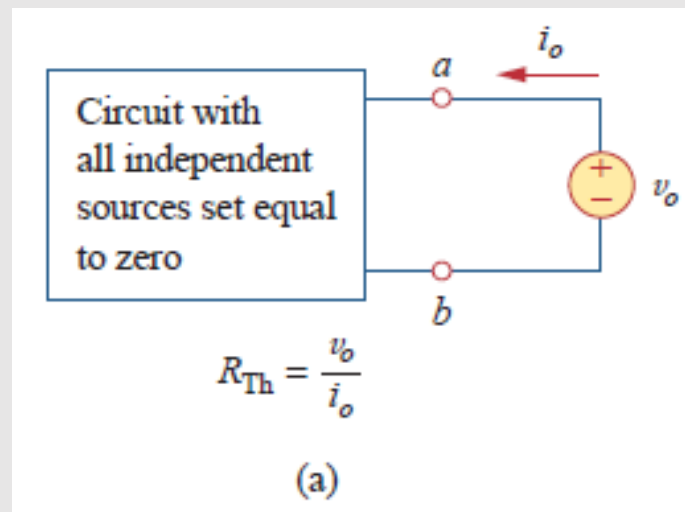


$$R_{Th} = R_{in}$$

(b)

Thevenin's Theorem

■ **CASE 2** If the network has dependent sources, we turn off all independent sources. As with superposition, dependent sources are not to be turned off because they are controlled by circuit variables. We apply a voltage source v_o at terminals a and b and determine the resulting current i_o . Then $R_{Th} = v_o/i_o$, as shown in Fig. 4.25(a). Alternatively, we may insert a current source i_o at terminals a - b as shown in Fig. 4.25(b) and find the terminal voltage v_o . Again $R_{Th} = v_o/i_o$. Either of the two approaches will give the same result. In either approach we may assume any value of v_o and i_o . For example, we may use $v_o = 1$ V or $i_o = 1$ A, or even use unspecified values of v_o or i_o .

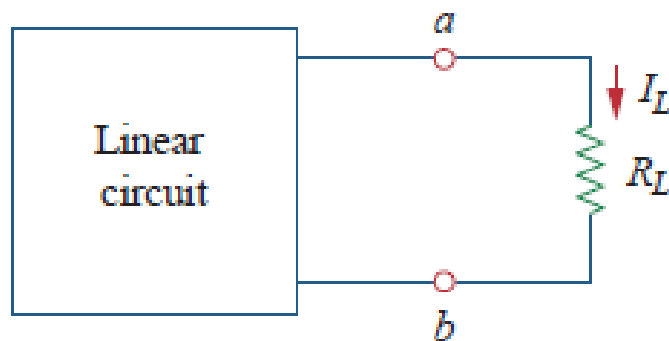


Thevenin's Theorem

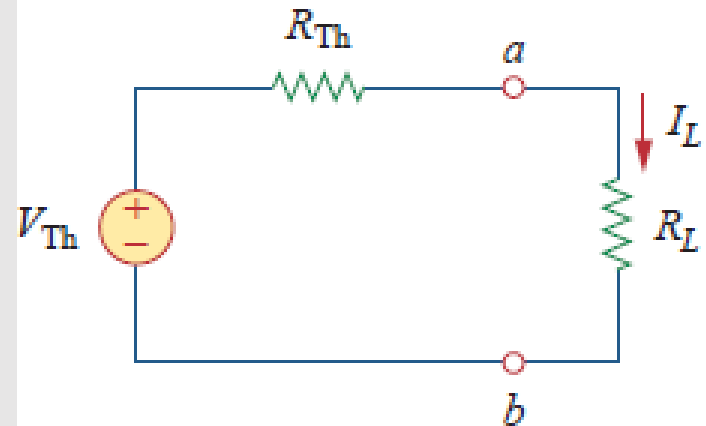
Thevenin's theorem is very important in circuit analysis. It helps simplify a circuit. A large circuit may be replaced by a single independent voltage source and a single resistor. This replacement technique is a powerful tool in circuit design.

$$I_L = \frac{V_{Th}}{R_{Th} + R_L}$$

$$V_L = R_L I_L = \frac{R_L}{R_{Th} + R_L} V_{Th}$$



(a)



(b)

Thevenin's Theorem

Find the Thevenin equivalent of the circuit in Fig. 4.31 at terminals a - b .

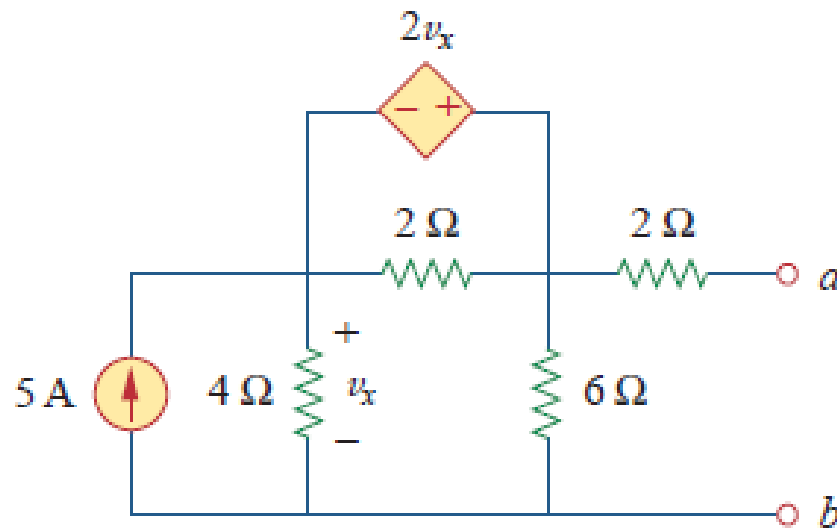
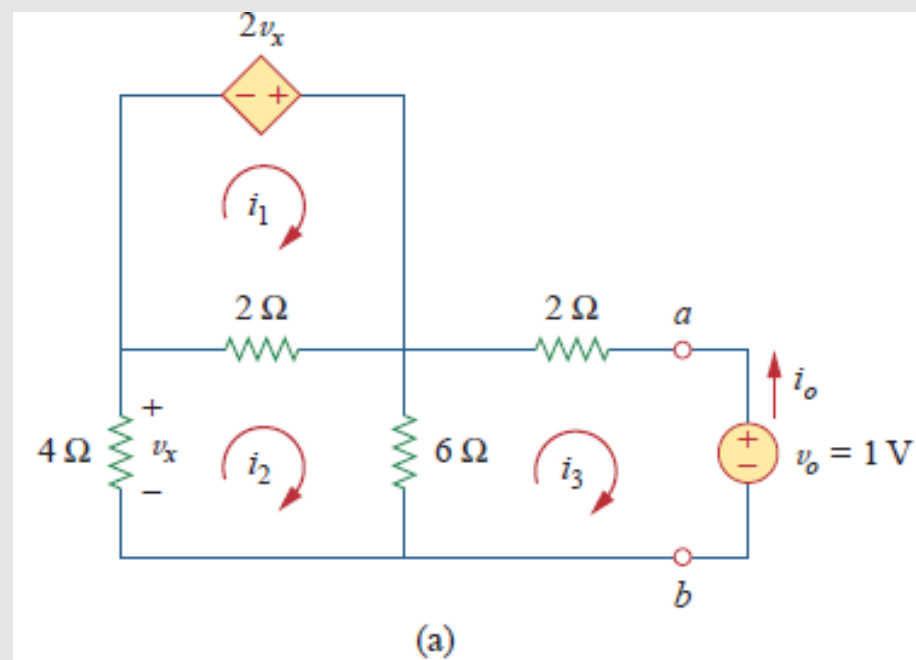


Figure 4.31
For Example 4.9.

Solution:

This circuit contains a dependent source, unlike the circuit in the previous example. To find R_{Th} , we set the independent source equal to zero but leave the dependent source alone. Because of the presence of the dependent source, however, we excite the network with a voltage source v_o connected to the terminals as indicated in Fig. 4.32(a). We may set $v_o = 1\text{ V}$ to ease calculation, since the circuit is linear. Our goal is to find the current i_o through the terminals, and then obtain $R_{Th} = 1/i_o$. (Alternatively, we may insert a 1-A current source, find the corresponding voltage v_o , and obtain $R_{Th} = v_o/1$.)



Applying mesh analysis to loop 1 in the circuit of Fig. 4.32(a) results in

$$-2v_x + 2(i_1 - i_2) = 0 \quad \text{or} \quad v_x = i_1 - i_2$$

But $-4i_2 = v_x = i_1 - i_2$; hence,

$$i_1 = -3i_2 \quad (4.9.1)$$

For loops 2 and 3, applying KVL produces

$$4i_2 + 2(i_2 - i_1) + 6(i_2 - i_3) = 0 \quad (4.9.2)$$

$$6(i_3 - i_2) + 2i_3 + 1 = 0 \quad (4.9.3)$$

Solving these equations gives

$$i_3 = -\frac{1}{6} \text{ A}$$

But $i_o = -i_3 = 1/6 \text{ A}$. Hence,

$$R_{\text{Th}} = \frac{1 \text{ V}}{i_o} = 6 \Omega$$

To get V_{Th} , we find v_{oc} in the circuit of Fig. 4.32(b). Applying mesh analysis, we get

$$i_1 = 5 \quad (4.9.4)$$

$$-2v_x + 2(i_3 - i_2) = 0 \quad \Rightarrow \quad v_x = i_3 - i_2 \quad (4.9.5)$$

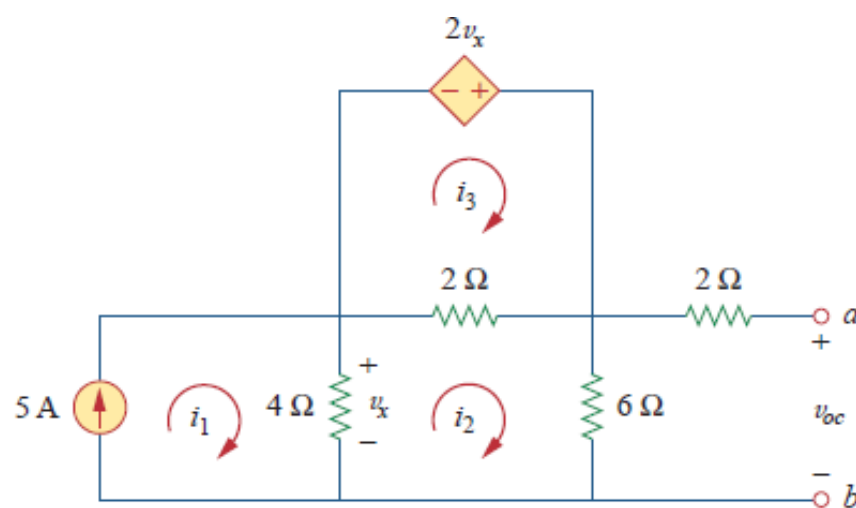
$$4(i_2 - i_1) + 2(i_2 - i_3) + 6i_2 = 0$$

or

$$12i_2 - 4i_1 - 2i_3 = 0 \quad (4.9.6)$$

But $4(i_1 - i_2) = v_x$. Solving these equations leads to $i_2 = 10/3$. Hence,

$$V_{Th} = v_{oc} = 6i_2 = 20 \text{ V}$$



(b)

Self Try

Find the Thevenin equivalent circuit of the circuit in Fig. 4.34 to the left of the terminals.

Answer: $V_{Th} = 5.333 \text{ V}$, $R_{Th} = 444.4 \text{ m}\Omega$.

Practice Problem 4.9

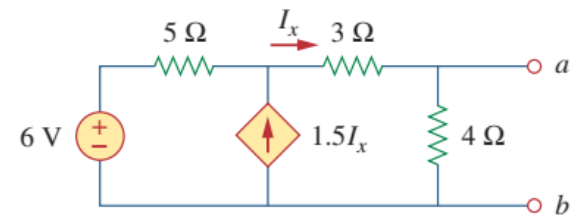


Figure 4.34

For Practice Prob. 4.9.

Norton's Theorem

Norton's theorem states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a current source I_N in parallel with a resistor R_N , where I_N is the short-circuit current through the terminals and R_N is the input or equivalent resistance at the terminals when the independent sources are turned off.

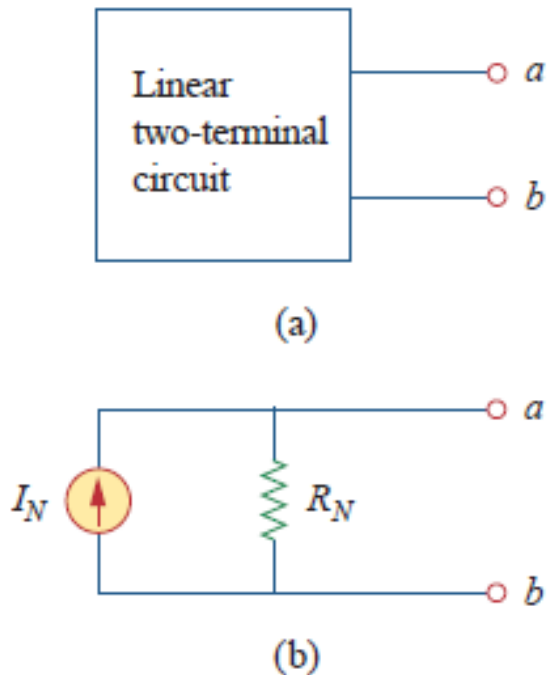


Figure 4.37

(a) Original circuit, (b) Norton equivalent circuit.

Norton's Theorem

The Thevenin and Norton equivalent circuits are related by a source transformation.

$$I_N = \frac{V_{Th}}{R_{Th}}$$

$$V_{Th} = v_{oc} \quad (4.12a)$$

$$I_N = i_{sc} \quad (4.12b)$$

$$R_{Th} = \frac{v_{oc}}{i_{sc}} = R_N \quad (4.12c)$$

the open-circuit and short-circuit tests are sufficient to find any Thevenin or Norton equivalent, of a circuit which contains at least one independent source.

Norton's Theorem

Using Norton's theorem, find i_x and v_x of the circuit in Fig. 4.43 at terminals a - b .

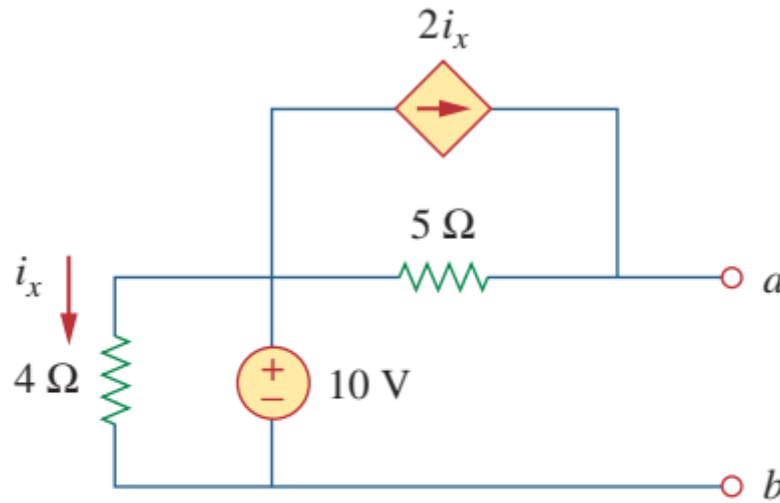


Figure 4.43
For Example 4.12.

Norton's Theorem

Solution:

To find R_N , we set the independent voltage source equal to zero and connect a voltage source of $v_o = 1$ V (or any unspecified voltage v_o) to the terminals. We obtain the circuit in Fig. 4.44(a). We ignore the $4\text{-}\Omega$ resistor because it is short-circuited. Also due to the short circuit, the $5\text{-}\Omega$ resistor, the voltage source, and the dependent current source are all in parallel. Hence, $i_x = 0$. At node a , $i_o = \frac{1v}{5\Omega} = 0.2$ A, and

$$R_N = \frac{v_o}{i_o} = \frac{1}{0.2} = 5\ \Omega$$

To find I_N , we short-circuit terminals a and b and find the current i_{sc} , as indicated in Fig. 4.44(b). Note from this figure that the $4\text{-}\Omega$ resistor, the 10-V voltage source, the $5\text{-}\Omega$ resistor, and the dependent current source are all in parallel. Hence,

$$i_x = \frac{10}{4} = 2.5\text{ A}$$

At node a , KCL gives

$$i_{sc} = \frac{10}{5} + 2i_x = 2 + 2(2.5) = 7\text{ A}$$

Thus,

$$I_N = 7\text{ A}$$

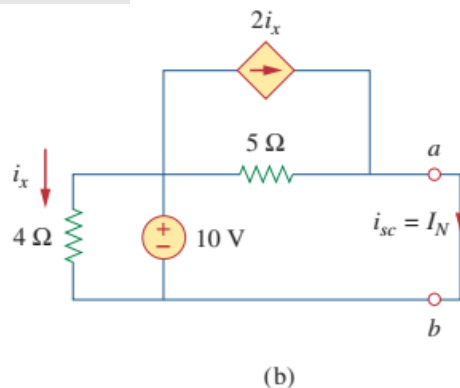
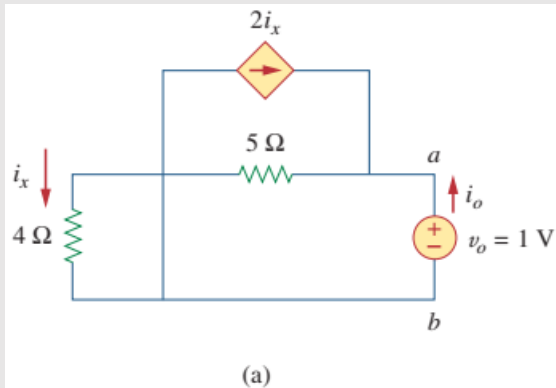


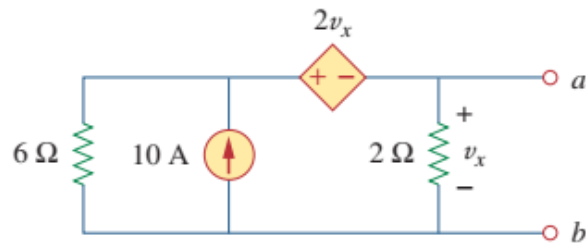
Figure 4.44

For Example 4.12: (a) finding R_N , (b) finding I_N .

Norton's Theorem

Practice Problem 4.12

Find the Norton equivalent circuit of the circuit in Fig. 4.45 at terminals a - b .



Answer: $R_N = 1 \Omega$, $I_N = 10 \text{ A}$.

Figure 4.45

For Practice Prob. 4.12.

Capacitors and Inductors

Capacitors and Inductors

Unlike resistors, which dissipate energy, capacitors and inductors do not dissipate but store energy, which can be retrieved at a later time. For this reason, capacitors and inductors are called storage elements.

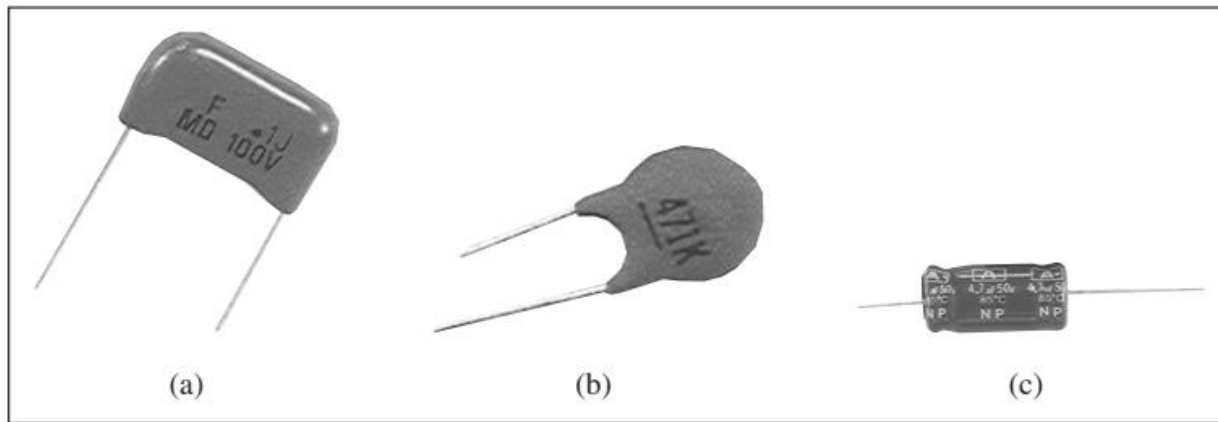


Figure 6.4

Fixed capacitors: (a) polyester capacitor, (b) ceramic capacitor, (c) electrolytic capacitor.
Courtesy of Tech America.

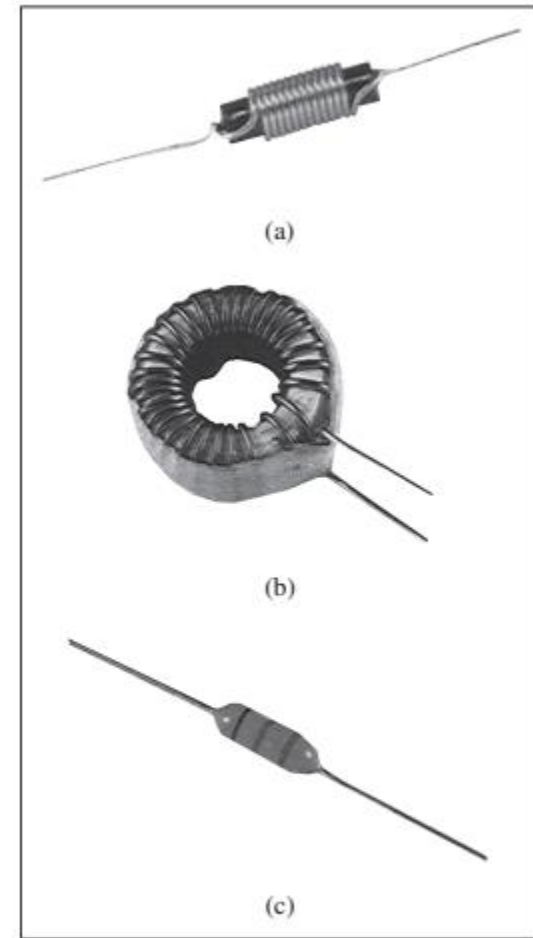


Figure 6.22

Various types of inductors: (a) solenoidal wound inductor, (b) toroidal inductor, (c) chip inductor.

Courtesy of Tech America.

Capacitor and Capacitance

- A capacitor is a passive element designed to store energy in its electric field.
- A capacitor consists of two conducting plates separated by an insulator (or dielectric)
- When a voltage source v is connected to the capacitor, as in Fig. 6.2, the source deposits a positive charge $+q$ on one plate and a negative charge $-q$ on the other.

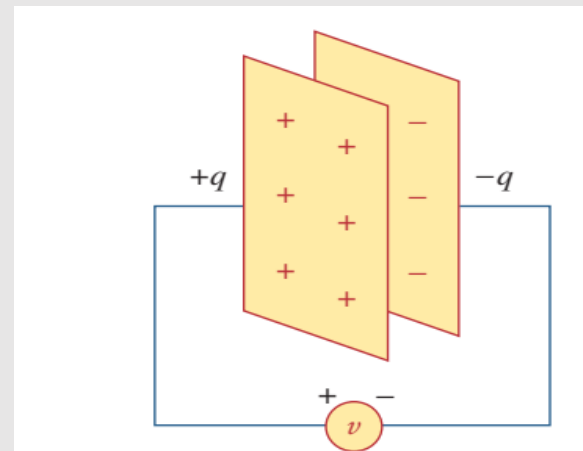


Figure 6.2

A capacitor with applied voltage v .

Capacitor and Capacitance (Contd.)

- The amount of charge stored, represented by q , is directly proportional to the applied voltage v so that

$$q \propto v$$

$$\text{or,} \quad q = Cv$$

Where C , the constant of proportionality, is known as the capacitance of the capacitor. The unit of capacitance is the farad (F), in honor of the English physicist Michael Faraday.

Capacitance is the ratio of the charge on one plate of a capacitor to the voltage difference between the two plates, measured in farads (F). 1 farad 1 coulomb/volt.

Capacitor and Capacitance (Contd.)

- Although the capacitance C of a capacitor is the ratio of the charge q per plate to the applied voltage it does not depend on q or It depends on the physical dimensions of the capacitor.
- For example, for the parallel-plate capacitor shown in Fig. 6.1, the capacitance is given by

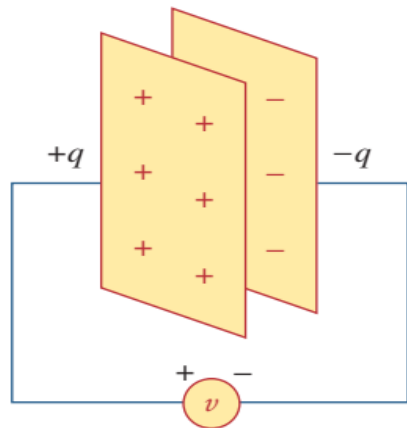


Figure 6.2

A capacitor with applied voltage v .

$$C = \frac{\epsilon A}{d}$$

where A is the surface area of each plate, d is the distance between the plates, and ϵ is the permittivity of the dielectric material between the plates.

Capacitor and Capacitance (Contd.)

- To obtain the **current-voltage relationship** of the capacitor, we take the derivative of both sides of

$$q = Cv$$

$$\text{Since, } i = \frac{dq}{dt}$$

$$\text{Thus } i = C \frac{dv}{dt}$$

- According to Eq, for a capacitor to carry current, its voltage must vary with time. **Hence, for constant voltage, $i = 0$.**

Capacitor and Capacitance (Contd.)

- The **voltage-current relation** of the capacitor can be obtained by integrating both sides of Eq. (6.4). We get

$$v(t) = \frac{1}{C} \int_{t_0}^t i(\tau) d\tau + v(t_0)$$

Where $v(t_0) = \frac{q(t_0)}{C}$ is the voltage across the capacitor at time t_0 .

Above Eqn shows **that capacitor voltage depends on the past history of the capacitor current. Hence, the capacitor has memory—a property** that is often exploited.

Capacitor and Capacitance (Contd.)

The instantaneous power delivered to the capacitor is

$$p = vi = Cv \frac{dv}{dt}$$

The energy stored in the capacitor is therefore

$$w = \int_{-\infty}^t p(\tau) d\tau = C \int_{-\infty}^t v \frac{dv}{d\tau} d\tau = C \int_{v(-\infty)}^{v(t)} v dv = \frac{1}{2} C v^2 \Big|_{v(-\infty)}^{v(t)}$$

We note that $v(-\infty)=0$, because the capacitor was uncharged at $t = -\infty$. Thus,

$$w = \frac{1}{2} C v^2 = \frac{q^2}{2C}$$

Some Important Property of Capacitors

- Note from Eq. (6.4) that when the voltage across a capacitor is not changing with time (i.e., dc voltage), the current through the capacitor is zero. Thus, A capacitor is an open circuit to dc. However, if a battery (dc voltage) is connected across a capacitor, the capacitor charges.
- The voltage on the capacitor must be continuous. The voltage on a capacitor cannot change abruptly. Conversely, the current through a capacitor can change instantaneously.

A discontinuous change in voltage requires an infinite current, which is physically impossible.

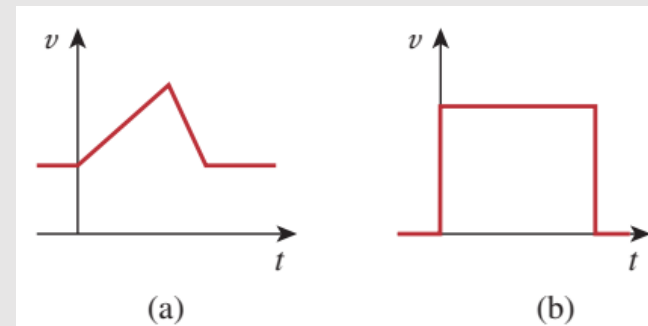


Figure 6.7

Voltage across a capacitor: (a) allowed, (b) not allowable; an abrupt change is not possible.

Example 6.4

Determine the current through a $200\text{-}\mu\text{F}$ capacitor whose voltage is shown in Fig. 6.9.

Example 6.4

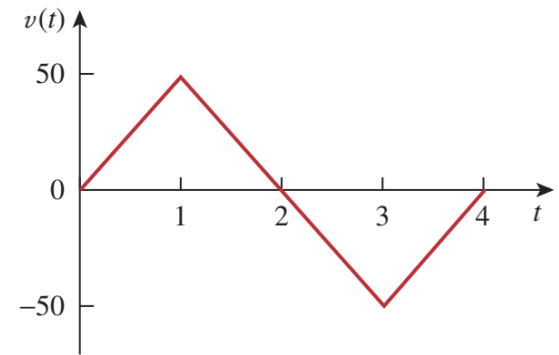


Figure 6.9
For Example 6.4.

Solution:

The voltage waveform can be described mathematically as

$$v(t) = \begin{cases} 50t \text{ V} & 0 < t < 1 \\ 100 - 50t \text{ V} & 1 < t < 3 \\ -200 + 50t \text{ V} & 3 < t < 4 \\ 0 & \text{otherwise} \end{cases}$$

Since $i = C dv/dt$ and $C = 200 \mu\text{F}$, we take the derivative of v to obtain

$$i(t) = 200 \times 10^{-6} \times \begin{cases} 50 & 0 < t < 1 \\ -50 & 1 < t < 3 \\ 50 & 3 < t < 4 \\ 0 & \text{otherwise} \end{cases}$$
$$= \begin{cases} 10 \text{ mA} & 0 < t < 1 \\ -10 \text{ mA} & 1 < t < 3 \\ 10 \text{ mA} & 3 < t < 4 \\ 0 & \text{otherwise} \end{cases}$$

Thus the current waveform is as shown in Fig. 6.10.

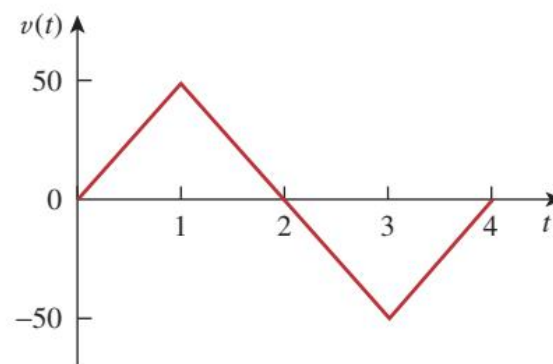


Figure 6.9
For Example 6.4.

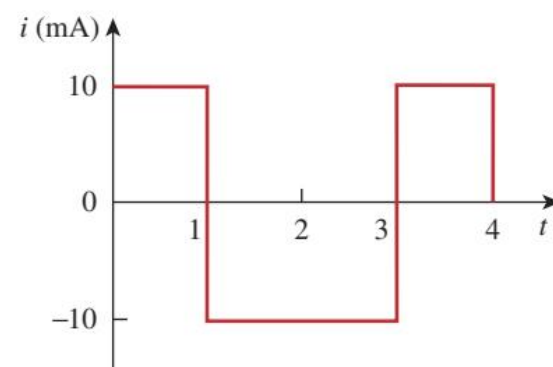


Figure 6.10
For Example 6.4.

Exercise 6.4

An initially uncharged 1-mF capacitor has the current shown in Fig. 6.11 across it. Calculate the voltage across it at $t = 2$ ms and $t = 5$ ms.

Answer: 100 mV, 400 mV.

Practice Problem 6.4

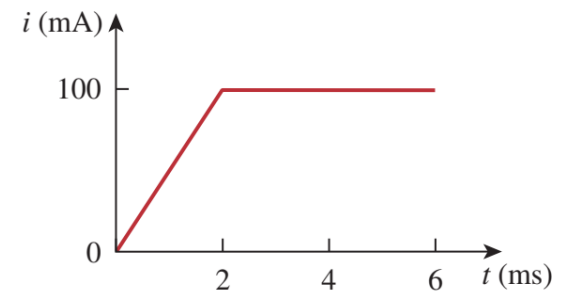
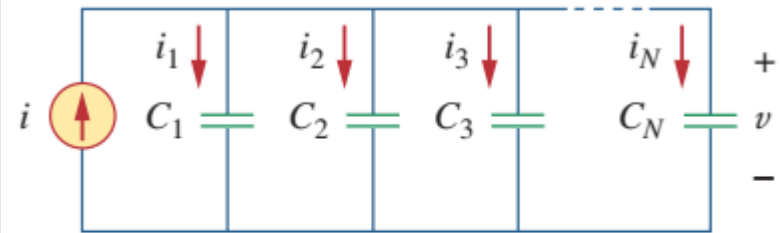


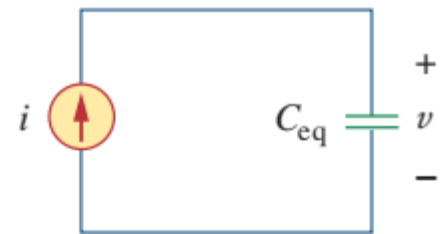
Figure 6.11

For Practice Prob. 6.4.

Parallel Capacitors



(a)



(b)

$$C_{eq} = C_1 + C_2 + C_3 + \cdots + C_N \quad (6.13)$$

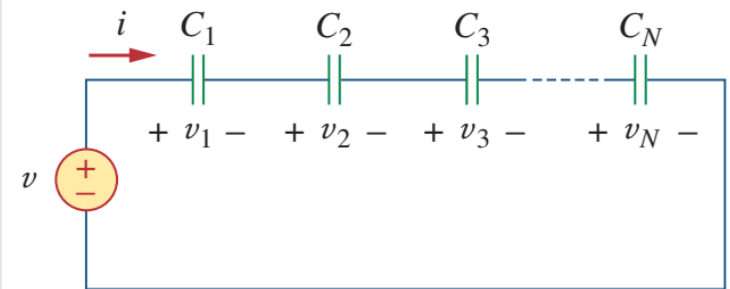
The **equivalent capacitance** of N parallel-connected capacitors is the sum of the individual capacitances.

Series Capacitors

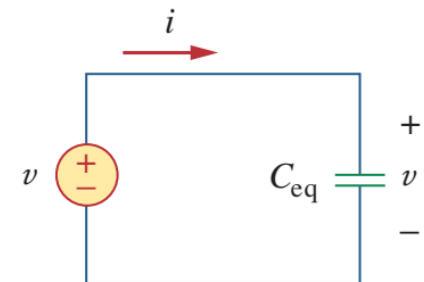
The **equivalent capacitance** of series-connected capacitors is the reciprocal of the sum of the reciprocals of the individual capacitances.

where

$$\frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots + \frac{1}{C_N}$$



(a)



(b)

Figure 6.15

(a) Series-connected N capacitors,
(b) equivalent circuit for the series capacitor.

Example 6.7

For the circuit in Fig. 6.18, find the voltage across each capacitor.

Example 6.7

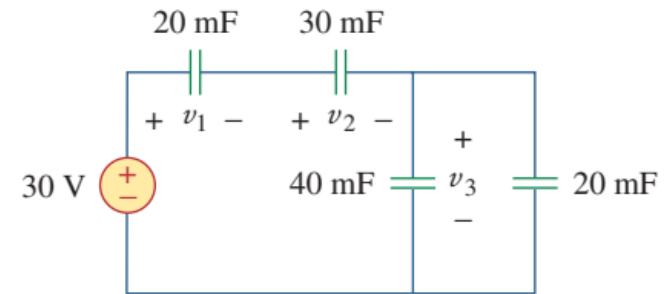


Figure 6.18
For Example 6.7.

Solution:

We first find the equivalent capacitance C_{eq} , shown in Fig. 6.19. The two parallel capacitors in Fig. 6.18 can be combined to get $40 + 20 = 60$ mF. This 60-mF capacitor is in series with the 20-mF and 30-mF capacitors. Thus,

$$C_{\text{eq}} = \frac{1}{\frac{1}{60} + \frac{1}{30} + \frac{1}{20}} \text{ mF} = 10 \text{ mF}$$

The total charge is

$$q = C_{\text{eq}}v = 10 \times 10^{-3} \times 30 = 0.3 \text{ C}$$

This is the charge on the 20-mF and 30-mF capacitors, because they are in series with the 30-V source. (A crude way to see this is to imagine that charge acts like current, since $i = dq/dt$.) Therefore,

$$v_1 = \frac{q}{C_1} = \frac{0.3}{20 \times 10^{-3}} = 15 \text{ V} \quad v_2 = \frac{q}{C_2} = \frac{0.3}{30 \times 10^{-3}} = 10 \text{ V}$$

Having determined v_1 and v_2 , we now use KVL to determine v_3 by

$$v_3 = 30 - v_1 - v_2 = 5 \text{ V}$$

Alternatively, since the 40-mF and 20-mF capacitors are in parallel, they have the same voltage v_3 and their combined capacitance is $40 + 20 = 60$ mF. This combined capacitance is in series with the 20-mF and 30-mF capacitors and consequently has the same charge on it. Hence,

$$v_3 = \frac{q}{60 \text{ mF}} = \frac{0.3}{60 \times 10^{-3}} = 5 \text{ V}$$

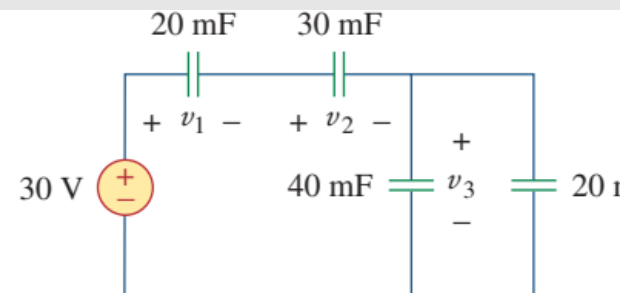


Figure 6.18
For Example 6.7.

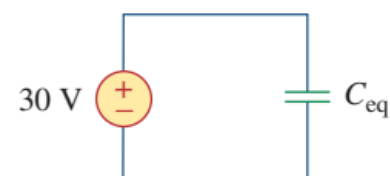


Figure 6.19
Equivalent circuit for Fig. 6.18.

Inductors

- An inductor is a passive element designed to store energy in its magnetic field. An inductor consists of a coil of conducting wire.
- If current is allowed to pass through an inductor, it is found that the voltage across the inductor is directly proportional to the time rate of change of the current.

$$v_L = L \frac{di_L}{dt}$$

- where L is the constant of proportionality called the inductance of the inductor. The unit of inductance is the henry (H).

Inductance is the property whereby an inductor exhibits opposition to the change of current flowing through it, measured in henrys (H).

Inductors (Contd.)

- The **current-voltage relationship** is obtained by integration

$$di = \frac{1}{L} v dt$$

Integrating gives

$$i = \frac{1}{L} \int_{-\infty}^t v(\tau) d\tau$$

or

$$i = \frac{1}{L} \int_{t_0}^t v(\tau) d\tau + i(t_0)$$

- Where $i(t_0)$ is the total current for an $-\infty < t < t_0$ and $i(-\infty) = 0$

Inductors (Contd.)

The inductor is designed to **store energy** in its magnetic field.

The energy stored can be obtained as, the power delivered to the inductor is

$$p = vi = \left(L \frac{di}{dt} \right) i$$

The energy stored is

$$\begin{aligned} w &= \int_{-\infty}^t p(\tau) d\tau = L \int_{-\infty}^t \frac{di}{d\tau} i d\tau \\ &= L \int_{-\infty}^t i di = \frac{1}{2} Li^2(t) - \frac{1}{2} Li^2(-\infty) \end{aligned}$$

Since $i(-\infty) = 0$,

$$w = \frac{1}{2} Li^2$$

Some Important Properties of Inductors

- Note that from $v_L = L \frac{di_L}{dt}$, the **voltage across an inductor is zero when the current is constant**. Thus, An inductor acts like a short circuit to dc.
- An important property of the inductor is its **opposition to the change in current flowing through it**. The current through an inductor cannot change instantaneously.

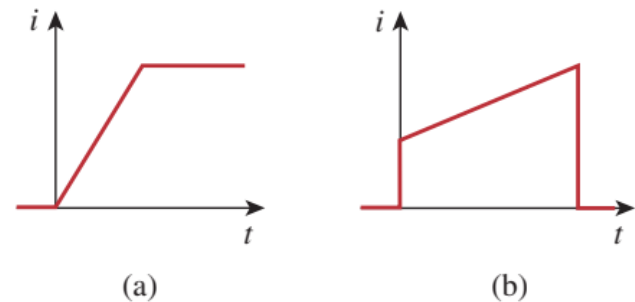


Figure 6.25

Current through an inductor: (a) allowed, (b) not allowable; an abrupt change is not possible.

Series Inductors

The **equivalent inductance** of series-connected inductors is the sum of the individual inductances.

$$L_{eq} = L_1 + L_2 + L_3 + \cdots + L_N$$

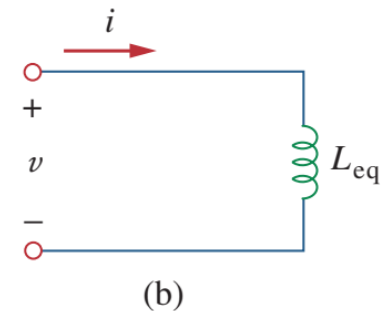
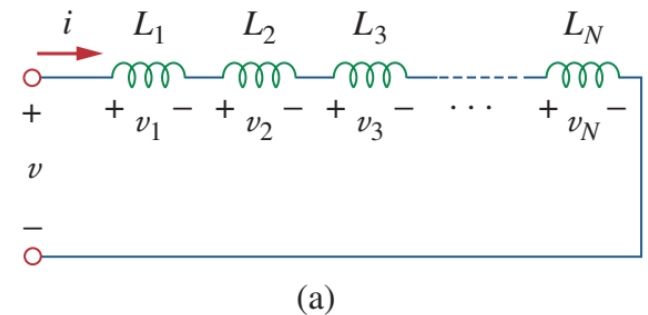


Figure 6.29

(a) A series connection of N inductors,
(b) equivalent circuit for the series inductors.

Parallel Inductors

The **equivalent inductance** of parallel inductors is the reciprocal of the sum of the reciprocals of the individual inductances.

$$\frac{1}{L_{\text{eq}}} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} + \cdots + \frac{1}{L_N}$$

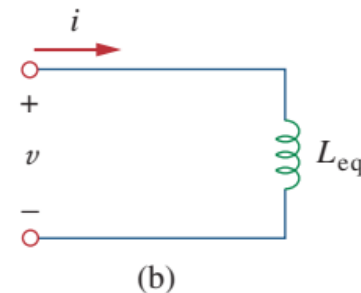
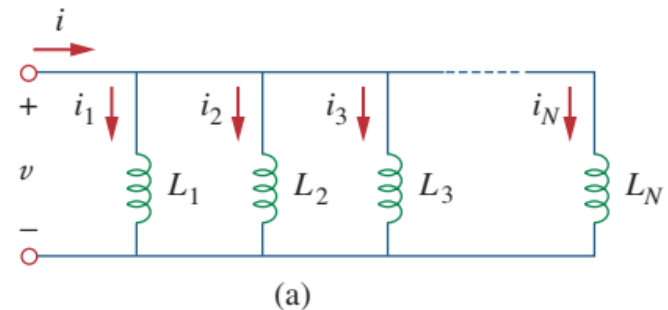


Figure 6.30

(a) A parallel connection of N inductors,
(b) equivalent circuit for the parallel inductors.

Example 6.12

For the circuit in Fig. 6.33, $i(t) = 4(2 - e^{-10t})$ mA. If $i_2(0) = -1$ mA, find: (a) $i_1(0)$; (b) $v(t)$, $v_1(t)$, and $v_2(t)$; (c) $i_1(t)$ and $i_2(t)$.

Example 6.12

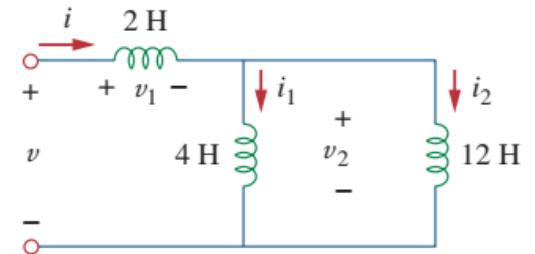


Figure 6.33

For Example 6.12.

Solution:

(a) From $i(t) = 4(2 - e^{-10t})$ mA, $i(0) = 4(2 - 1) = 4$ mA. Since $i = i_1 + i_2$,

$$i_1(0) = i(0) - i_2(0) = 4 - (-1) = 5 \text{ mA}$$

(b) The equivalent inductance is

$$L_{\text{eq}} = 2 + 4 \parallel 12 = 2 + 3 = 5 \text{ H}$$

Thus,

$$v(t) = L_{\text{eq}} \frac{di}{dt} = 5(4)(-1)(-10)e^{-10t} \text{ mV} = 200e^{-10t} \text{ mV}$$

and

$$v_1(t) = 2 \frac{di}{dt} = 2(-4)(-10)e^{-10t} \text{ mV} = 80e^{-10t} \text{ mV}$$

Since $v = v_1 + v_2$,

$$v_2(t) = v(t) - v_1(t) = 120e^{-10t} \text{ mV}$$

(c) The current i_1 is obtained as

$$\begin{aligned} i_1(t) &= \frac{1}{4} \int_0^t v_2 dt + i_1(0) = \frac{120}{4} \int_0^t e^{-10t} dt + 5 \text{ mA} \\ &= -3e^{-10t} \Big|_0^t + 5 \text{ mA} = -3e^{-10t} + 3 + 5 = 8 - 3e^{-10t} \text{ mA} \end{aligned}$$

Similarly,

$$\begin{aligned} i_2(t) &= \frac{1}{12} \int_0^t v_2 dt + i_2(0) = \frac{120}{12} \int_0^t e^{-10t} dt - 1 \text{ mA} \\ &= -e^{-10t} \Big|_0^t - 1 \text{ mA} = -e^{-10t} + 1 - 1 = -e^{-10t} \text{ mA} \end{aligned}$$

Note that $i_1(t) + i_2(t) = i(t)$.

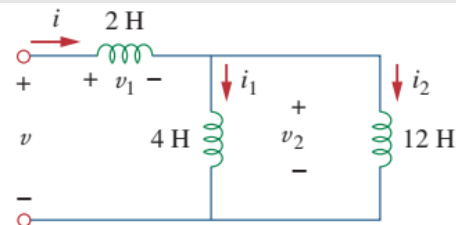


Figure 6.33

For Example 6.12.



Thank You